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Analysis of a queue-length-dependent vacation queue with bulk service, N-policy, set-up time and cost optimization

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ABSTRACT

Due to the extensive applications of bulk service vacation queues in manufacturing industries, inventory systems, wireless sensor networks for deducing energy consumption etc., in this article, we analyze the steady-state behavior of an infinite-buffer group arrival bulk service queue with vacation scenario, set-up time and N-threshold policy. Here the customers arrive according to the compound Poisson process and the server originates the service process with minimum 'a' customers and can give service to maximum 'b' customers at a time. We adopt batch-size-dependent service time as well as queue-length-dependent vacation duration which improve the system's performance significantly. The N-threshold policy is proposed to awaken the server from a vacation/dormant state where the service station starts the set-up procedure after the accumulation of pre-decided 'N' customers. Using the supplementary variable technique, firstly, we derive the set of system equations in the steady-state. After that, we obtain the bivariate probability generating functions (pgfs) of queue content and size of the departing batch, the queue content and type of vacation taken by the server at vacation completion epoch and also the single pgf of queue content at the end of set-up time. We extract the joint distribution from those generating functions using the roots method and derive a simple algebraic relation between the probabilities at departure and arbitrary epoch. We also provide assorted numerical results to validate our proposed methodology and obtained theoretical results. The impact of the system parameters on the performance measures is presented through tables and graphs. Finally, a cost optimization function is provided for the benefit of system designers.

1. Introduction

Over the last decade, bulk service queues with or without batch-size-dependent service have received extensive attention among researchers due to their growing use as mathematical models of road traffic flow, production and manufacturing assembly lines, web servers, the movement of aircraft passengers, wireless sensor networks (WSNs), packet switching network systems, data processing etc. In recent times, WSNs have been used widely in several many fields, for example, in traffic management, also to collect data regarding the environmental conditions (such as pollution levels, pressure, temperature, humidity and wind movement), in military defense systems (to detect troops, intruders, equipment and coordinate service to fight), in the health care system where the sensor monitors the health of the human body etc. For more theoretical and analytical development along with applications on batch service queueing systems, see Yu and Alfa [1], Barbhuiya and Gupta [2], Samanta and Nandi [3] and references therein.

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In order to improve the quality of service (QoS) and efficient use of the different nodes in WSNs, several mechanisms have been proposed such as congestion control algorithms, data aggregation, multi-path routing etc., see Fasolo et al. [4], Geng et al. [5]. However, the batch forwarding mechanism is a significant tool that mainly collects the data from the sensor nodes and sends these in terms of batches to the base station. In the batch forwarding mechanism, one can impose the batch-size-dependent service time (see Pradhan and Gupta [6], Banerjee et al. [7,8]), where the transmission time depends on the batch size and this type of mechanism reduces the delay time and improve the system consistency and productivity substantially.

In several queueing systems, after completion of the transmission/service of a batch, the service station often inspects the size of the waiting line. If the number of waiting packets is less than the pre-decided threshold value, then it leaves the service area temporarily. After a random amount of time, again the server comes back to the service area, and if an adequate number of data or packets is available then it commences the service process. Otherwise, the server waits until the required number of customers (packets, data) come or again may go for another vacation depending on the pre-decided vacation policy under which it is operating. This type of scenario frequently occurs in manufacturing industries (where the machine's breakdown, repairing or maintenance can be taken into account as vacation period), transportation, computer and telecommunication system, health-care facility systems, banks etc. Also, the queueing models with vacation have huge applications in WSNs to reduce energy consumption with preserving the QoS and improving the lifetime of the battery. In this context, see Sun et al. [9] and Sajwan et al. [10] and references therein for different mechanisms. Among these, the sleeping mode technique is the most effective mechanism to save energy in WSNs at the present time. Liu et al. [11] and Yang et al. [12] have discussed the strategies on random sleeping policies for energy efficiency in WSNs. Several researchers have put their attention towards the discrete-time queues also, see Samanta et al. [13–15], Nandy and Pradhan [16]. In these analyses, vacation time was fixed irrespective of queue length.

During the vacation period, if the server takes more time to return to 'on state' then sometimes the arrived customers (data, packets) have to wait a long time in the queue for transmission or getting some other services. This may lead the customer's impatient as well as increase the loss ratio. To overcome this, the queue-length-dependent vacation duration (vacation time depends on queue length at the start of vacation) is an important tool that reduces the queue size, waiting time, loss ratio and improves the system's efficiency. Using this concept, Gupta et al. [17] have analyzed finite-buffer $M/G_n^{(a,b)}/1$ queue under both single and multiple vacations, where they have derived the joint distribution of queue and server content as well as the joint distribution of queue and type of vacation taken by the server at different time epochs. Further, Tamrakar and Banerjee [18] have studied the same model by considering the infinite waiting space. It is worthy mentioning that the above literature did not consider any kind of threshold policies and set-up time.

To avoid the frequent switch-over between off state and on state, there are several control policies available such as N-policy, T-policy, D-policy, F-policy etc. Among these, N-policy is the most traditional control policy which is frequently used by several researchers due to its broad applications in flexible manufacturing systems, several service systems, inventory systems etc. In queueing systems with N-policy, the server is turned on from 'off state' when there are at least 'N' customers waiting in the queue. Recently, some researchers have focused on finite-/infinite-buffer queueing models with N-policy such as Barbhuiya et al. [19,20], Kempa and Marjasz [21], Yang [22]. Under N-policy, when the server is turned on, it takes some amount of time to prepare itself before serving the waiting customers, which is known as set-up time. The queueing models with N-policy and set-up time have energetic applications in WSNs for saving energy consumption, see Guo et al. [23], Merit et al. [24]. There is little few literature available where the authors considered the bulk arrival and bulk service together with vacation, threshold policy and set-up time, for example, Reddy et al. [25], Jeyakumar and Arumuganathan [26], and Haridass and Arumuganathan [27]. In the above literature, the authors focused only on the derivation of the probability generating function (pgf) of queue content at arbitrary epoch without complete extraction process. Recently, Pradhan and Karan [28] have investigated $M/G_n^{(a,b)}/1$ under multiple vacation and server breakdown, where they have obtained the joint pgf of the queue and server content at departure epoch and have derived the complete joint distribution at departure epoch.

Recently, Karan et al. [29] have analyzed $M/G_n^{(a,b)}/1$ vacation queueing model with N-policy, set-up time, and batch-size-dependent service time in which customers arrive individually according to a Poisson process. In the model, we have proposed, in this paper, customers arrive in groups of random sizes, i.e., in accordance with compound Poisson process. This makes the analysis quite complex to analyze from modeling and computational perspective as selected customers from an arriving batch may not be always served together within the same batch. This article includes queue-length-dependent vacation time while the previous did not which is another major difference. This also increases the complexity and difficulty of the derivation of bivariate pgfs and associated other results. The advantage of queue-length-dependent vacation time is that it reduces the vacation duration, hence the excessive delay in WSN can be minimized and service process may be speeded up, which eventually reduces energy consumption as well as system cost. The recent work of this paper includes system cost optimization through the concept of busy and idle cycle length which was lacking in the previous article. It also focuses the detailed discussion about the sensitivity analysis of different parameters on performance measures. In particular, the effect of arrival rate for various service and vacation time distributions is elaborately discussed in the current concerned model.

All the above literature reveals that only a few works have been done on bulk service with N-policy, and set-up time without the complete extraction of both queue and server content distribution (except our previous work [29]). Sometimes, it is necessary to know both queue and server content distribution to control the traffic in several situations. For example, in WSNs, at the relay nodes, the packets arrive in bulk and are transmitted in batches with a minimum and maximum threshold. To reduce the congestion, it is needed to speed up the transmission rate which is only possible when both the loads at the queue level as well as the server level are known to us. In such situations, only the queue content distribution may not be sufficient for reducing the congestion. Also, for minimizing the operating cost, it is required to determine the minimum and maximum service capacity, for which the

server content distribution is required. Furthermore, the batch-size-dependent service and queue-length-dependent vacation policy together play an important role in reducing the queue length as well as the customer's waiting time and also directly affect the minimization of cost.

These motivate us to model a batch arrival and batch-size-dependent service queue with queue-length-dependent both single and multiple vacations, N-threshold policy, and set-up time. To the best of the authors' knowledge, there is no literature available on the joint queue and server content distribution and also other marginal distributions along with performance measures and optimization of the system's operating cost. The major contributions of current work are twofold. Firstly, due to many characteristic features such as queue-length-dependent single as well as multiple vacation, batch-size-dependent service time, set-up time and N-threshold policy, the construction of transition probability matrix (TPM) and derivation of queue and server content distribution through embedded Markov chain technique (EMCT) is tedious and hard task, if not impossible. However, the use of supplementary variable technique (SVT) overcomes this complexity, where the remaining service time, vacation time and set-up time are taken into account as the supplementary variables. We sketch three major real-life applications of our concerned model. Use of SVT leads to the system equations in the form of difference–differential equations. Using the Laplace–Stieltjes transform and double generating function technique, we derive the explicit form of the joint pgf of the joint distribution of queue content and number of customers with the departing batch. We also derive the bivariate pgf of queue content and type of vacation taken by the server and the pgf of only queue content at the end of vacation and set-up time, respectively. Secondly, the extraction mechanism of the corresponding probability distribution is provided elegantly. A simple relation between departure and arbitrary epoch probabilities is built up and from this, the significant marginal distributions are formulated. For the practitioners, some performances measures are also reported which can be readily achieved using the relevant marginal distributions. From the practical point of view, we develop a cost function and provide the optimal values of some parameters. Also, the effect of different parameters on the system's performances are presented through tables and graphs which may be fruitful to the system designers from applications perspective.

The remaining portion of this paper is arranged as follows: Section 2 contains practical applications of the concerned model, while Section 3 provides a detailed description of the model. The underlying Markov chain, the steady-state governing system equations and the complete procedure of the derivation of pgfs are presented in Section 4. In Section 5, we describe the process of determination of unknowns appearing in the pgfs with the help of some derived lemmas and the extraction of corresponding probabilities. The algebraic relation between the departure and arbitrary epoch probabilities is provided in Section 6. In Section 7, we derive some relevant marginal distributions along with performance measures. Finally, the numerical example and cost function are displayed in Section 8 followed by the conclusions and future scopes of the work.

2. Practical application of the concerned model

Here we point out a application of our concerned model in a major real-life circumstance.

(i) Wireless sensor networks: In recent times, several significant applications of queues in WSNs have been pointed out by Xu et al. [30], Stankevich et al. [31] etc. In WSNs, the sensor nodes collect the data from their vicinity area and sent the data to the base station. However, sometimes the sensor nodes are placed outside the communication range of the base station. In such cases, the other nodes are placed between the base station and sensor nodes which are generally called relay nodes. The main work of relay nodes is to collect the data from the sensor nodes and send towards the final destination (i.e., destined to the base station). To improve the quality of service (QoS) and efficient use of the relay nodes, many researchers have proposed several mechanisms such as congestion control algorithms, Hull et al. [32], multi-path routing, Geng et al. [5], data aggregation, see Fasolo et al. [4]. Here we can use a batch forwarding mechanism which is similar as data aggregation. In the batch forwarding mechanism, first, the relay nodes collect the data from the sensor nodes and put them into a large container called the batch message and send only these batches to the base station. The batch forwarding mechanism has some advantages as compared to data aggregation mechanism, firstly the aim of the conventional data aggregation method is to collect the data and process those collected data in the networks, and finally send to the base station. On the contrary, batch forwarding does not process the collected data, simply collects data from the sensor nodes and sends it to the base station. However in some cases, we do not want data aggregation in the network. Suppose one would like to track an object or see the materialistic or spatial distribution of the sensory readings data over time and/or field. In such a case, we require every sensory reading to the base station. Secondly, the data aggregation mechanism works on the same type of collected data. However in many cases, the different types of data (for example, temperature, pressure, vibration etc.) may be required. In this case, the data aggregation mechanism cannot be applied. Hence batch forwarding mechanism is an applicable resolution. Also the batch forwarding mechanism improves the delivery ratio and reduces the delay performance in WSNs. In batch forwarding mechanism, we can impose the batch-size-dependent service time policy which is one of the major features of the concerned model. In batch-size-dependent service, the transmission time dynamically varies with the batch size, which reduces the delay time and improves the system's productivity and consistency substantially.

In our model, the vacation time (single or multiple) can exactly be used as sleeping modes (single or multiple) which is the most cost-effective mechanism to reduce the energy consumption in WSNs system. Also the implementation of queue-length-dependent vacation policy in WSNs reduces vacation time, and hence more batches can be transferred within a stipulated time which eventually minimizes system cost.

For more applications of the proposed models in various real-life circumstances, readers may visit the link https://github.com/KaranPrasenjit/Supplement.

3. Model description

We provide a detailed description of the concerned queueing model below.

- **Arrival process:** Here the customers (data, packets, jobs) arrive in batches/groups according to a time homogeneous classical Poisson process with arrival rate λ. The sizes of arrival batches $\{X_i\}_{i=1}^{\infty}$ are independent and identically distributed (i.i.d) and composed as a generic random variable X having the probability mass function (pmf) $Pr(X = j) = g_j$, $j = 1, 2, \ldots$, finite mean $E[X] = \bar{g}$ and the associated pgf $G(z) = \sum_{j=1}^{\infty} g_j z^j$. In practical scenarios, most of the time the size of arrival batches is finite. For this reason, during the computation, we consider the finite size of arrival batches.
- **Service rule:** The arrived customers are served in batches on the basis of first-come first-service (FCFS) by a single server according to the general bulk service rule which is usually known as (a, b) rule. The server initiates the service process when the number of waiting customers is at least 'a'. It (server) can provide the service to a maximum 'b' ($\geq a$) customers at a time. If the number of waiting customers is 'r' ($a \leq r \leq b$), then all the customers are considered for service and if the queue size (waiting customer) is greater than 'b', then it takes only the first 'b' customers to provide the required service and the remaining customers wait for the next round of service. Newly arrived customers are not allowed to enter the ongoing service process even if there is free space.
- **Service process:** In several real-life phenomena, the service/processing times do not follow any particular conventional probability distribution. Keeping this in mind, here we assume that the service/processing times follow the general distribution so that a large class of distributions can be used for service and also the service time depends on the ongoing serving batch size. This special service mechanism reduces the congestion as well as speeds up the embedded process. Let Y_r be the random variable (RV) denoting the service time of a batch of size 'r' ($a \leq r \leq b$) with probability density function (pdf) $s_r(t)$, distribution function (df) $S_r(t)$, the Laplace–Stieltjes transform (L.S.T) $\tilde{S}_r(\theta)$. The mean service time of batch of size 'r' is $\frac{1}{\mu_r} = s_r = -\tilde{S}_r^{(1)}(0)$, where $\tilde{S}_r^{(n)}(0)$ means the nth order derivative of $\tilde{S}_r(\theta)$ calculated at $\theta = 0$.
- **Vacation process and N-policy:** Whenever the minimum threshold value is not met for initiation of service, the server leaves the system for a random amount of time generally known as vacation time. This concerned model is analyzed for two different vacation policies such as single vacation and multiple vacations. It is very important to mention that the system can operate only with either single or multiple vacations throughout. The change of vacation policy after the completion of one or several service process is not permitted. Here the system equations and all the results are provided together for single as well as multiple vacations which can be separately extracted using the indicator function δ_p defined as follows:
$$\delta_p = \begin{cases} 0, & \text{when the system goes through a single vacation} \\ 1, & \text{when the system goes through multiple vacations.} \end{cases}$$
The results of this queue can be obtained by substituting $\delta_p = 0$ in the case of a single vacation and for multiple vacations by substituting $\delta_p = 1$.
- **Single vacation rule:** After finishing off the service of a batch, if the server finds that the queue size is greater than or equal to 'a' then it recommences the service process according to (a, b) rule, otherwise it goes to a vacation. The vacation time depends on the queue size 'k' ($0 \leq k \leq a - 1$) at the initiation of the vacation period. At the termination of this vacation, if the aggregated customers are greater than or equal to a pre-decided threshold value, say, 'N' ($\geq b$), then the server starts the set-up immediately, otherwise it waits until the threshold value (which is known as dormant state) is achieved.
- **Multiple vacations rule:** As soon as the service of a batch is completed, the server inspects the waiting line, if the number of customers is less than the minimum threshold value 'a', instantly it becomes unavailable for random vacation periods which depends on the customers 'k' ($0 \leq k \leq a-1$). Otherwise, it continues the service process according to the above mentioned (a, b) rule. At the vacation completion epoch, if the queue length is at least the pre-decided value, say 'N' ($\geq b$), the server starts the set-up else it again goes to another vacation and the vacation time depends on the present queue size 'k' ($0 \leq k \leq N - 1$) at the beginning of the vacation period. The random length of the 'k'-th type vacation periods are represented by the generic variable $\Psi^{[k]}$. We assume that all vacations are i.i.d with pdf $\psi^{[k]}(t)$, df $\Psi^{[k]}(t)$, L.S.T $\tilde{\Psi}^{[k]}(\theta)$ and finite mean $E(\Psi^{[k]}) = \bar{\psi}^{[k]}$= -$\tilde{\Psi}^{[k](1)}(0) = \dfrac{1}{\eta_k}$.
- **Set-up process:** Upon returning from a vacation/dormant state, whenever the server finds that 'N' or more customers in the queue, then it starts the set-up of a random length (which is represented by the RV R) before providing the service. After completing the set-up, the server immediately initiates the service process with 'b' number of customers and the rest of the customers wait for the next round of service. The pdf and df of the RV R are $r(t)$ and $R(t)$, respectively. The L.S.T of $R(t)$ is denoted by $\tilde{R}(\theta)$ and the mean value of set-up time is given by $E(R) = \bar{r} = -\tilde{R}^{(1)}(0) = \dfrac{1}{\nu}$.
- **Traffic intensity:** The traffic intensity of the system is $\rho = \frac{\lambda \bar{g}}{b\mu_b} < 1$, which ensures the stability of the system.

The concerned model is presented graphically in Figs. 1 and 2 for single and multiple vacations, respectively.

Remark 1. It is worth to mention here that we assume the arrival, service, vacation and set-up process to be mutually independent and eventually lead to no correlation among them.

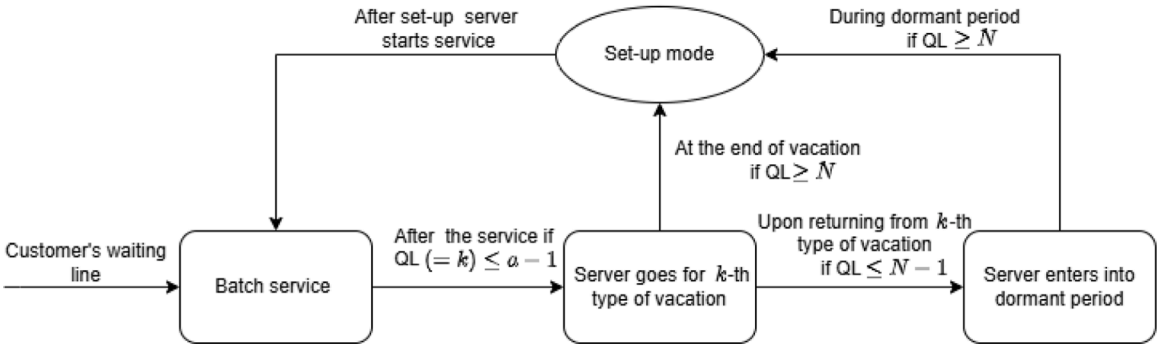

Fig. 1. Pictorial representation of the considered single vacation model.

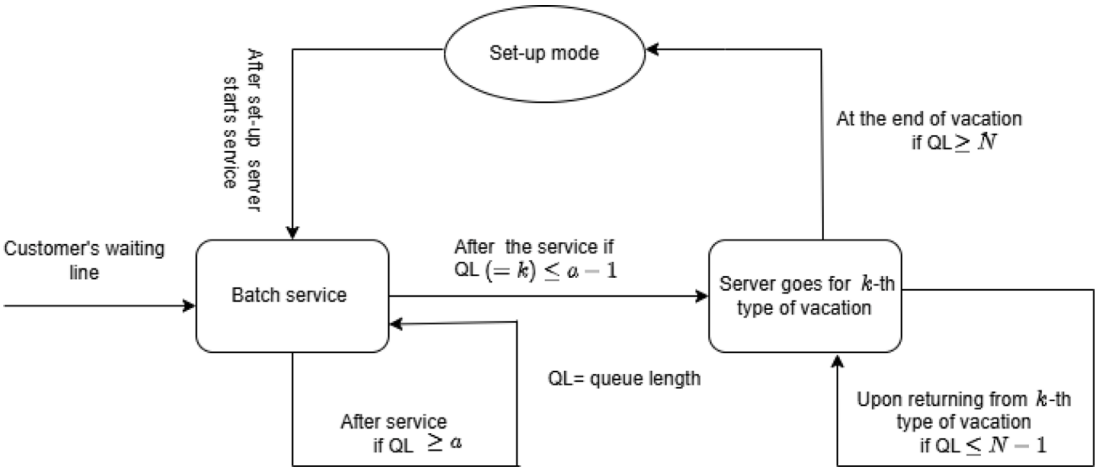

Fig. 2. Pictorial representation of the considered multiple vacation model.

4. Derivation of pgfs at service, vacation and set-up completion epoch

In this section, first we describe the underlying Markov chain of the concerned model by considering the remaining service, vacation and set-up time as supplementary variables. Then using SVT, we construct the steady-state governing equations of the system in terms of difference–differential equations by observing the system at two consecutive arbitrary time epochs t and $t+dt$, where dt measures the very small time. We derive some results in the lemma form using which finally, we obtain the bivariate pgf of queue and server content at service completion/departure epoch, the bivariate pgf of queue content and type of vacation at vacation completion epoch and the single pgf of queue content at the end of set-up time.

4.1. The associated Markov chain

At any arbitrary time t, the system can be described by the process $(\eta_p(t), \widehat{N}_q(t), \widehat{N}_s(t), \widehat{U}(t), \widehat{A}^{[k]}(t), \widehat{T}(t))$, where $\eta_p(t)$ represents the state of the server with $\eta_p(t) = 0$, 1, 2 or 3 according to either the server is on the dormant state or vacation period or busy state or set-up mode, respectively. $\widehat{N}_q(t)$ denotes the number of customers waiting in the queue for service, $\widehat{N}_s(t)$ represents the number of customers in the server when the server is busy. $\widehat{U}(t)$ denotes the remaining service time of batch in service, $\widehat{A}^{[k]}(t)$ corresponds to the remaining kth vacation time of the server, and $\widehat{T}(t)$ designates the remaining set-up time of the server. Then $\{(\eta_p(t), \widehat{N}_q(t), \widehat{N}_s(t), \widehat{U}(t), \widehat{A}^{[k]}(t), \widehat{T}(t))\}$ represents the Markov process of the concerned model with the state spaces $\{(n,0) : n = 0,\dots,a-1;\ (n,r,u) : n \geq 0,\ r = a,\dots,b,\ u \geq 0;\ (n,k,u) : n \geq k,\ k = 0,\dots,a-1,\ u \geq 0;\ (n,u) : n \geq N,\ u \geq 0\}$ for single vacation and $\{(n,r,u) : n \geq 0,\ r = a,\dots,b,\ u \geq 0;\ (n,k,u) : n \geq k,\ k = 0,\dots,N-1,\ u \geq 0;\ (n,u) : n \geq N,\ u \geq 0\}$ for multiple vacations.

Further, we define the following probabilities at time t as:

$$\begin{aligned}
v_n(t) &= Pr\{\widehat{N}_q(t) = n,\ \widehat{N}_s(t) = 0,\ \eta_p(t) = 0\}, \quad 0 \leq n \leq N-1,\\
p_{n,r}(x,t)dx &= Pr\{\widehat{N}_q(t) = n,\ \widehat{N}_s(t) = r,\ x < \widehat{U}(t) \leq x+dx,\ \eta_p(t) = 2\}, \quad x \geq 0, \quad n \geq 0, \quad a \leq r \leq b,\\
q_n^{[k]}(x,t)dx &= Pr\{\widehat{N}_q(t) = n,\ \widehat{N}_s(t) = 0,\ x < \widehat{A}^{[k]}(t) \leq x+dx,\ \eta_p(t) = 1\}, \quad x \geq 0, \quad n \geq k,
\end{aligned}$$

$$0 \le k \le a-1 \text{ (for single vacation) or } 0 \le k \le N-1 \text{ (for multiple vacations)},$$

$$r_n(x,t)dx = Pr\{\widehat{N}_q(t)=n,\ \widehat{N}_s(t)=0,\ x<\widehat{T}(t)\le x+dx,\ \eta_p(t)=3\},\quad x\ge 0,\quad n\ge N.$$

Since our main concern is to obtain the stationary joint distributions, we first define the following limiting probabilities as :

$$\begin{aligned}
\upsilon_n &= \lim_{t\to\infty}\upsilon_n(t),\quad 0\le n\le N-1,\\
p_{n,r}(x) &= \lim_{t\to\infty}p_{n,r}(x,t),\quad n\ge 0,\ a\le r\le b,\\
q_n^{[k]}(x) &= \lim_{t\to\infty}q_n^{[k]}(x,t),\quad n\ge k,\ 0\le k\le a-1 \text{ or } 0\le k\le N-1,\\
r_n(x) &= \lim_{t\to\infty}r_n(x,t),\quad n\ge N,
\end{aligned}$$

4.2. The system equations

Relating the states of the system at two consecutive time epochs t and $t+dt$ and using SVT, and also considering the limiting case, after some simplification, we obtain the following steady-state system equations in the difference–differential form:

$$0 = (1-\delta_p)\left(-\lambda\upsilon_0 + q_0^{[0]}(0)\right), \tag{1}$$

$$0 = (1-\delta_p)\left(-\lambda\upsilon_n + \sum_{k=0}^{min(a-1,n)} q_n^{[k]}(0) + \lambda\sum_{j=1}^{n} g_j\upsilon_{n-j}\right),\quad 1\le n\le N-1, \tag{2}$$

The above two equations correspond to the dormant states in case of a single vacation and vanishes for multiple vacations. More precisely, these represent the transition among the states during of single vacation and dormant periods.

$$-\frac{d}{dx}p_{0,r}(x) = -\lambda p_{0,r}(x) + \sum_{m=a}^{b} p_{r,m}(0)s_r(x),\quad a\le r\le b, \tag{3}$$

$$-\frac{d}{dx}p_{n,r}(x) = -\lambda p_{n,r}(x) + \lambda\sum_{j=1}^{n} g_j p_{n-j,r}(x),\quad n\ge 1,\quad a\le r\le b-1, \tag{4}$$

$$-\frac{d}{dx}p_{n,b}(x) = -\lambda p_{n,b}(x) + \sum_{m=a}^{b} p_{n+b,m}(0)s_b(x) + \lambda\sum_{j=1}^{n} g_j p_{n-j,b}(x),\quad 1\le n\le N-b-1, \tag{5}$$

$$-\frac{d}{dx}p_{n,b}(x) = -\lambda p_{n,b}(x) + \sum_{m=a}^{b} p_{n+b,m}(0)s_b(x) + \lambda\sum_{j=1}^{n} g_j p_{n-j,b}(x) + r_{n+b}(0)s_b(x),\quad n\ge N-b, \tag{6}$$

The Eqs. (3)–(6) correspond to the busy states i.e., the server is busy with some customers while $n\ge 0$ customers are waiting in the queue. In these equations, the transitions occur among the busy states and set-up states in which service times and arrival group sizes play a significant role.

$$-\frac{d}{dx}r_N(x) = -\lambda r_N(x) + \left(\sum_{k=0}^{a-1} q_N^{[k]}(0) + \delta_p\sum_{k=a}^{N-1} q_N^{[k]}(0)\right)r(x) + \lambda(1-\delta_p)\sum_{j=0}^{N-1} g_{N-j}\upsilon_j r(x), \tag{7}$$

$$\begin{aligned}
-\frac{d}{dx}r_n(x) = -\lambda r_n(x) &+ \left(\sum_{k=0}^{a-1} q_n^{[k]}(0) + \delta_p\sum_{k=a}^{N-1} q_n^{[k]}(0)\right)r(x) + \lambda(1-\delta_p)\sum_{j=0}^{N-1} g_{N-j}\upsilon_j r(x)\\
&+\lambda\sum_{j=N}^{n-1} g_{n-j}r_j(x),\quad n\ge N+1,
\end{aligned} \tag{8}$$

The Eqs. (7)–(8) represent the system equations whenever the server is in set-up state and the number of customers present in the queue is either N or more than N along with dormant states and vacation states.

$$-\frac{d}{dx}q_0^{[0]}(x) = -\lambda q_0^{[0]}(x) + \left(\sum_{m=a}^{b} p_{0,m}(0) + \delta_p q_0^{[0]}(0)\right)\psi^{[0]}(x), \tag{9}$$

$$-\frac{d}{dx}q_k^{[k]}(x) = -\lambda q_k^{[k]}(x) + \left(\sum_{r=a}^{b} p_{k,r}(0) + \delta_p\sum_{m=0}^{k} q_k^{[m]}(0)\right)\psi^{[k]}(x),\quad 1\le k\le a-1, \tag{10}$$

$$-\frac{d}{dx}q_k^{[k]}(x) = -\lambda q_k^{[k]}(x) + \sum_{m=0}^{k} q_k^{[m]}(0)\psi^{[k]}(x),\quad a\le k\le N-1, \tag{11}$$

$$-\frac{d}{dx}q_n^{[k]}(x) = -\lambda q_n^{[k]}(x) + \lambda\sum_{j=1}^{n} g_j q_{n-j}^{[k]}(x),\quad 0\le k\le a-1,\ n\ge k+1, \tag{12}$$

$$-\frac{d}{dx}q_n^{[k]}(x) = -\lambda q_n^{[k]}(x) + \lambda\sum_{j=1}^{n} g_j q_{n-j}^{[k]}(x),\quad a\le k\le N-1,\ n\ge k+1, \tag{13}$$

Finally, Eqs. (9)–(13) designate the system equations whenever the server is in vacation state. The term $\psi^{[k]}(x)$ clearly indicates the dependence of vacation time on the queue length.

Remark 2. In the case of a single vacation, k varies from 0 to $a-1$, whereas for multiple vacations, k varies from 0 to $N-1$ and so the Eqs. (11) and (13) occur only for multiple vacations cases.

4.3. The bivariate pgfs at service/vacation completion epoch

As our main focal point is to obtain the bivariate pgfs of the queue and server content at the departure epoch and queue content and type of vacation at vacation completion epoch from (1)–(13), for this, first we define the following probabilities at departure epoch:

$$p^{+}_{n,r} = Pr\{n \text{ customers in the queue at departure epoch and } r \text{ customers with the departing batch}\},$$
$$n \geq 0, \quad a \leq r \leq b, \tag{14}$$

$$p^{+}_{n} = Pr\{n \text{ customers in waiting line at departure epoch of a batch}\} = \sum_{r=a}^{b} p^{+}_{n,r}, \quad n \geq 0, \tag{15}$$

$$q^{[k]+}_{n} = Pr\{n \text{ customers in the queue at the end of } k\text{th type of vacation time}\}, \quad n \geq k,$$
$$0 \leq k \leq a-1 \text{ or } 0 \leq k \leq N-1, \tag{16}$$

$$q^{+}_{n} = Pr\{n \text{ customers in the queue at the end of vacation period}\}$$
$$= (1-\delta_p) \sum_{k=0}^{min(n,a-1)} q^{[k]+}_{n} + \delta_p \sum_{k=0}^{min(n,N-1)} q^{[k]+}_{n}, \quad n \geq 0, \tag{17}$$

$$r^{+}_{n} = Pr\{n \text{ customers in the queue at the end of set-up time}\}, \quad n \geq N. \tag{18}$$

The corresponding pgfs are defined as:

$$P(z,y,\theta) = \sum_{n=0}^{\infty}\sum_{r=a}^{b} \tilde{P}_{n,r}(\theta) z^n y^r, \quad |z| \leq 1, \quad |y| \leq 1, \tag{19}$$

$$P^{+}(z,y) = \sum_{n=0}^{\infty}\sum_{r=a}^{b} p^{+}_{n,r} z^n y^r, \quad |z| \leq 1, \quad |y| \leq 1, \tag{20}$$

$$P^{+}(z,1) = \sum_{n=0}^{\infty}\sum_{r=a}^{b} p^{+}_{n,r} z^n = \sum_{n=0}^{\infty} p^{+}_{n} z^n = P^{+}(z), \quad |z| \leq 1, \tag{21}$$

$$R^{+}(z) = \sum_{n=N}^{\infty} r^{+}_{n} z^n, \quad |z| \leq 1, \tag{22}$$

$$Q^{+}(z,y) = \sum_{k=0}^{a-1}\sum_{n=k}^{\infty} q^{[k]+}_{n} z^n y^k + \delta_p \sum_{k=a}^{N-1}\sum_{n=k}^{\infty} q^{[k]+}_{n} z^n y^k = \sum_{n=0}^{\infty}\sum_{k=0}^{min(n,a-1)} q^{[k]+}_{n} z^n y^k +$$
$$\delta_p \sum_{n=a}^{\infty}\sum_{k=a}^{min(n,N-1)} q^{[k]+}_{n} z^n y^k, \quad |z| \leq 1, \ |y| \leq 1, \tag{23}$$

$$Q^{+}(z) = Q^{+}(z,1) = \sum_{n=0}^{\infty}\sum_{k=0}^{min(n,a-1)} q^{[k]+}_{n} z^n + \delta_p \sum_{n=a}^{\infty}\sum_{k=a}^{min(n,N-1)} q^{[k]+}_{n} z^n = \sum_{n=0}^{\infty} q^{+}_{n} z^n, \quad |z| \leq 1. \tag{24}$$

Now we define the following L.S.Ts and probabilities at arbitrary epoch:

$$\tilde{P}_{n,r}(\theta) = \int_0^{\infty} e^{-\theta x} p_{n,r}(x)dx, \quad n \geq 0, \ \tilde{S}_i(\theta) = \int_0^{\infty} e^{-\theta x} dS_i(x) = \int_0^{\infty} e^{-\theta x} s_i(x)dx, \quad a \leq r \leq b,$$
$$\tilde{\Psi}^{[k]}(\theta) = \int_0^{\infty} e^{-\theta x} d\Psi^{[k]}(x) = \int_0^{\infty} e^{-\theta x} \psi^{[k]}(x)dx, \quad 0 \leq k \leq a-1 \text{ or } 0 \leq k \leq N-1,$$
$$\tilde{R}(\theta) = \int_0^{\infty} e^{-\theta x} dR(x) = \int_0^{\infty} e^{-\theta x} r(x)dx, \quad \tilde{R}_n(\theta) = \int_0^{\infty} e^{-\theta x} r_n(x)dx, \quad n \geq 0,$$
$$\tilde{Q}^{[k]}_{n}(\theta) = \int_0^{\infty} e^{-\theta x} q^{[k]}_{n}(x)dx, \quad n \geq k, \quad 0 \leq k \leq N-1,$$
$$p_{n,r} = \tilde{P}_{n,r}(0) = \int_0^{\infty} p_{n,r}(x)dx, \quad a \leq r \leq b, \ n \geq 0, \quad r_n = \tilde{R}_n(0) = \int_0^{\infty} r_n(x)dx, \quad n \geq N,$$
$$q^{[k]}_{n} = \tilde{Q}^{[k]}_{n}(0) = \int_0^{\infty} q^{[k]}_{n}(x)dx, \quad n \geq k, \quad 0 \leq k \leq N-1.$$

Remark 3. Here $p_{n,r}$ denotes the probabilities that n customers are in the queue and r customers are with the server at arbitrary epoch and $q_n^{[k]}$ (r_n) are the probabilities that n customers are in the queue when the server is on kth type of vacation (set-up mode) at arbitrary epoch. Let q_n be the probability that n customers are waiting in the queue at arbitrary time epoch when the server is on vacation mode irrespective of the type of vacation and can be calculated as:

$$q_n = (1-\delta_p)\sum_{k=0}^{min(a-1,n)} q_n^{[k]} + \delta_p \sum_{k=0}^{min(N-1,n)} q_n^{[k]}, \quad n \geq 0,$$

In the similar way, $q_n(0)$ is the probability that n customers in the queue with no remaining vacation time and are given by:

$$q_n(0) = (1-\delta_p)\sum_{k=0}^{min(a-1,n)} q_n^{[k]}(0) + \delta_p \sum_{k=0}^{min(N-1,n)} q_n^{[k]}(0)$$

Before deriving the main results, we derive some results which will be applied during the derivation of main results.

Lemma 4.1. *The probabilities* $\{p_{n,r}^+,\ q_n^{[k]+},\ r_n^+\}$ *and* $\{p_{n,r}(0),\ q_n^{[k]}(0),\ r_n(0)\}$ *are connected by the following relations:*

$$p_{n,r}^+ = \frac{p_{n,r}(0)}{\Omega}, \quad n \geq 0,\ a \leq r \leq b, \tag{25}$$

$$q_n^{[k]+} = \frac{q_n^{[k]}(0)}{\Omega}, \quad n \geq k,\ 0 \leq k \leq a-1 \text{ or } 0 \leq k \leq N-1, \tag{26}$$

$$r_n^+ = \frac{r_n(0)}{\Omega}. \tag{27}$$

where $\Omega = \sum_{n=0}^{\infty}\sum_{r=a}^{b} p_{n,r}(0) + (1-\delta_p)\sum_{n=0}^{\infty}\sum_{k=0}^{min(n,a-1)} q_n^{[k]}(0) + \delta_p \sum_{n=0}^{\infty}\sum_{k=0}^{min(n,N-1)} q_n^{[k]}(0) + \sum_{n=N}^{\infty} r_n(0)$

It is noteworthy to mention that Eq. (25) designates the relation between joint queue and server length at service completion epoch with the joint probabilities of queue and server length when there is no service time left. Similar can be commented about (26) and (27) in terms of vacation and set-up, respectively. Moreover Ω^{-1} equals the mean of service completion or vacation completion or set-up completion per unit time i.e., mean departure rate either from busy state or vacation state or set-up state.

Proof. Here $p_{n,r}(0)$ denotes the joint probability that there are n customers in the queue and r customers with the server just prior to the service completion epoch i.e., no remaining service time.

Applying the conditional probability for $n \geq 0,\ a \leq r \leq b$, we have

$$\begin{aligned}
p_{n,r}^+ &= Pr(n \text{ customers in the queue and } r \text{ customers with server just prior to the service completion} \\
&\quad \text{epoch}|\text{total number of customers in the system just prior to the service completion or vacation} \\
&\quad \text{completion or set-up completion epoch}), \\
&= \lim_{t\to\infty} Pr\{\widehat{N}_q(t) = n,\ \widehat{N}_s(t) = r,\ \eta_p = 2,\ \widehat{U}(t) = 0 \mid \widehat{U}(t) = 0 \text{ or } \widehat{A}^{[k]}(t) = 0 \text{ or } \widehat{T}(t) = 0\} \\
&= \lim_{t\to\infty} \frac{Pr\{\widehat{N}_q(t) = n,\ \widehat{N}_s(t) = r,\ \eta_p = 2,\ \widehat{U}(t) = 0\}}{Pr\{\widehat{U}(t) = 0 \text{ or } \widehat{A}^{[k]}(t) = 0 \text{ or } \widehat{T}(t) = 0\}} \\
&= \frac{p_{n,r}(0)}{\Omega}
\end{aligned}$$

In the similar way, we can establish the relations (26) and (27). □

Lemma 4.2. *The probabilities* υ_n *and* $q_n(0)$ *are connected by*

$$\upsilon_n = \frac{1}{\lambda}\sum_{m=0}^{n} e_{n,m} q_m(0), \quad n = 0, \ldots, N-1. \tag{28}$$

where $e_{n,n-1} = g_1$, $e_{n,n} = 1$, $n = 0, \ldots, N-1$, $e_{n,i} = \sum_{j=1}^{n-1-i} e_{n-j,i} g_j + g_{n-i}$, $i = 0, \ldots, n-2$. *which represents a recursive relation among probabilities to construct the required number of customers with all possible combinations of* g_i*'s* $(i = 1, 2, \ldots)$.

Proof. In case of single vacation $\delta_p = 0$ and the Eqs. (1) and (2) appear only for single vacation.

Putting $\delta_p = 0$ in (1) and (2), we get

$$\lambda\upsilon_0 = q_0^{[0]}(0) = e_{0,0} q_0(0), \tag{29}$$

$$\lambda\upsilon_n = \sum_{k=0}^{min(a-1,n)} q_n^{[k]}(0) + \lambda\sum_{j=1}^{n} g_j \upsilon_{n-j}, \quad 1 \leq n \leq N-1, \tag{30}$$

Now substituting $n = 1$ in (30) and using (29), we obtain

$$\upsilon_1 = \frac{1}{\lambda}\sum_{m=0}^{1} e_{1,m} q_m(0)$$

By similar fashion, we acquire the required result. □

Lemma 4.3. *The value of* $\sum_{n=0}^{\infty}\sum_{r=a}^{b} p_{n,r}(0) + (1-\delta_p)\sum_{n=0}^{\infty}\sum_{k=0}^{min(n,a-1)} q_n^{[k]}(0) + \delta_p \sum_{n=0}^{\infty}\sum_{k=0}^{min(n,N-1)} q_n^{[k]}(0) + \sum_{n=N}^{\infty} r_n(0)$ *is given by*

$$\sum_{n=0}^{\infty}\sum_{r=a}^{b} p_{n,r}(0) + (1-\delta_p)\sum_{n=0}^{\infty}\sum_{k=0}^{min(n,a-1)} q_n^{[k]}(0) + \delta_p \sum_{n=0}^{\infty}\sum_{k=0}^{min(n,N-1)} q_n^{[k]}(0) + \sum_{n=N}^{\infty} r_n(0) = \frac{1-(1-\delta_p)\sum_{n=0}^{N-1}\upsilon_n}{\omega} \tag{31}$$

where $\omega = \sum_{n=0}^{a-1}\left(\bar{\Psi}^{[n]}p_n^+ + (1-\delta_p)q_n^+\sum_{j=n}^{N-1} e_{j,n}\sum_{i=N-j}^{\infty} g_i\bar{r} + \delta_p q_n^+\bar{\Psi}^{[n]}\right) + \sum_{n=a}^{b}\left(s_n p_n^+ + (1-\delta_p)q_n^+ \sum_{j=n}^{N-1} e_{j,n}\sum_{i=N-j}^{\infty} g_i\bar{r} + \delta_p\bar{\Psi}^{[n]}q_n^+\right) + \sum_{n=b+1}^{N-1}\left(s_b p_n^+ + (1-\delta_p)q_n^+\sum_{j=n}^{N-1} e_{j,n}\sum_{i=N-j}^{\infty} g_i\bar{r} + \delta_p\bar{\Psi}^{[n]}q_n^+\right) + \sum_{n=N}^{\infty}(s_b p_n^+ + \bar{r}q_n^+ + s_b r_n^+)$

Remark 4. The expression in ω is a relation among the number of customers left at the departure epoch, mean service, vacation, set-up time and possible cases of arrival group sizes. This quantity plays a significant role while computing joint probabilities at arbitrary epoch.

Proof. Multiplying Eqs. (3)–(13) by $e^{-\theta x}$ and integrating with respect to x over 0 to ∞, we obtain

$$(\lambda-\theta)\tilde{P}_{0,r}(\theta) = \sum_{m=a}^{b} p_{r,m}(0)\tilde{S}_r(\theta) - p_{0,r}(0), \quad a \le r \le b, \tag{32}$$

$$(\lambda-\theta)\tilde{P}_{n,r}(\theta) = \lambda\sum_{j=1}^{n} g_j\tilde{P}_{n-j,r}(\theta) - p_{n,r}(0), \quad a \le r \le b-1, \quad n \ge 1, \tag{33}$$

$$(\lambda-\theta)\tilde{P}_{n,b}(\theta) = \lambda\sum_{j=1}^{n} g_j\tilde{P}_{n-j}(\theta) + \sum_{m=a}^{b} p_{n+b,m}(0)\tilde{S}_b(\theta) - p_{n,b}(0), \quad 1 \le n \le N-b-1, \tag{34}$$

$$(\lambda-\theta)\tilde{P}_{n,b}(\theta) = \lambda\sum_{j=1}^{n} g_j\tilde{P}_{n-j,b}(\theta) + \sum_{m=a}^{b} p_{n+b,m}(0)\tilde{S}_b(\theta) + r_{n+b}(0)\tilde{S}_b(\theta) - p_{n,b}(0), \quad n \ge N-b, \tag{35}$$

$$(\lambda-\theta)\tilde{R}_N(\theta) = \left(\sum_{k=0}^{a-1} q_N^{[k]}(0) + \delta_p\sum_{k=a}^{N-1} q_N^{[k]}(0)\right)\tilde{R}(\theta) + \lambda(1-\delta_p)\sum_{j=0}^{N-1} g_{N-j}\upsilon_j\tilde{R}(\theta) - r_N(0), \tag{36}$$

$$(\lambda-\theta)\tilde{R}_n(\theta) = \left(\sum_{k=0}^{a-1} q_n^{[k]}(0) + \delta_p\sum_{k=a}^{N-1} q_n^{[k]}(0)\right)\tilde{R}(\theta) + \lambda(1-\delta_p)\sum_{j=0}^{N-1} g_{n-j}\upsilon_j\tilde{R}(\theta) + \lambda\sum_{j=N}^{n-1} g_{n-j}\tilde{R}_j(\theta) - r_n(0), \quad n \ge N+1, \tag{37}$$

$$(\lambda-\theta)\tilde{Q}_0^{[0]}(\theta) = \left(\sum_{m=a}^{b} p_{0,r}(0) + \delta_p q_0^{[0]}(0)\right)\tilde{\Psi}^{[0]}(\theta) - q_0^{[0]}(0), \tag{38}$$

$$(\lambda-\theta)\tilde{Q}_k^{[k]}(\theta) = \left(\sum_{r=a}^{b} p_{k,r}(0) + \delta_p\sum_{m=0}^{k} q_k^{[m]}(0)\right)\tilde{\Psi}^{[k]}(\theta) - q_k^{[k]}(0), \quad 1 \le k \le a-1, \tag{39}$$

$$(\lambda-\theta)\tilde{Q}_k^{[k]}(\theta) = \sum_{m=0}^{k} q_k^{[k]}(0)\tilde{\Psi}^{[k]}(\theta) - q_k^{[k]}(0), \quad a \le k \le N-1, \tag{40}$$

$$(\lambda-\theta)\tilde{Q}_n^{[k]}(\theta) = \lambda\sum_{j=1}^{n} g_j\tilde{Q}_{n-j}^{[k]}(\theta) - q_n^{[k]}(\theta), \quad 0 \le k \le a-1, \ n \ge k+1, \tag{41}$$

$$(\lambda-\theta)\tilde{Q}_n^{[k]}(\theta) = \lambda\sum_{j=1}^{n} g_j\tilde{Q}_{n-j}^{[k]}(\theta) - q_n^{[k]}(\theta), \quad a \le k \le N-1, \ n \ge k+1. \tag{42}$$

Using (28) in (36) and (37), summing over n from 0 to ∞ and r from a to b in Eqs. (32)–(42) and after some algebraic calculation, we obtain

$$\sum_{n=0}^{\infty}\sum_{r=a}^{b}\tilde{P}_{n,r}(\theta) + (1-\delta_p)\sum_{n=0}^{\infty}\sum_{k=0}^{min(n,a-1)}\tilde{Q}_n^{[k]}(\theta) + \delta_p\sum_{n=0}^{\infty}\sum_{k=0}^{min(n,N-1)}\tilde{Q}_n^{[k]}(\theta) + \sum_{n=N}^{\infty}\tilde{R}_n(\theta)$$
$$= \sum_{n=0}^{a-1}\sum_{r=a}^{b}\left(\frac{1-\tilde{\Psi}^{[n]}(\theta)}{\theta}\right)p_{n,r}(0) + (1-\delta_p)\sum_{n=0}^{N-1}\left(\frac{1-\sum_{j=n}^{N-1} e_{j,n}\sum_{i=N-j}^{\infty} g_i\tilde{R}(\theta)}{\theta}\right)q_n(0) +$$

$$\delta_p \sum_{n=0}^{a-1}\left(\frac{1-\tilde{\Psi}^{[n]}(\theta)}{\theta}\right)q_n(0)+\sum_{n=a}^{b}\sum_{r=a}^{b}\left(\frac{1-\tilde{S}_n(\theta)}{\theta}\right)p_{n,r}(0)+\delta_p\sum_{n=a}^{N-1}\left(\frac{1-\tilde{\Psi}^{[n]}(\theta)}{\theta}\right)q_n(0)+$$
$$\left(\frac{1-\tilde{S}_b(\theta)}{\theta}\right)\sum_{n=b+1}^{\infty}\sum_{r=a}^{b}p_{n,r}(0)+\left(\frac{1-\tilde{S}_b(\theta)}{\theta}\right)\sum_{n=N}^{\infty}r_n(0)+\left(\frac{1-\tilde{R}(\theta)}{\theta}\right)\sum_{n=N}^{\infty}q_n(0)$$

Taking the limit as $\theta \to 0$ on both the sides and applying L'Hospital's rule, additionally, using Lemma 4.1 and the normalizing condition $(1-\delta_p)\sum_{n=0}^{N-1}\nu_n+\sum_{n=0}^{\infty}\sum_{r=a}^{b}p_{n,r}+(1-\delta_p)\sum_{n=0}^{\infty}\sum_{k=0}^{min(a-1,n)}q_n^{[k]}+\delta_p\sum_{n=0}^{\infty}\sum_{k=0}^{min(N-1,n)}q_n^{[k]}+\sum_{n=N}^{\infty}r_n=1$, we achieve the desired result. □

Theorem 4.4. *The bivariate pgf of queue content and type of vacation taken by the server at the end of vacation time and the pgf of queue content at the end of set-up are given by*

$$Q^+(z,y) = \sum_{n=0}^{a-1}p_n^+z^ny^nM^{[n]}(z)+\delta_p\sum_{n=0}^{N-1}q_n^+z^ny^nM^{[n]}(z), \tag{43}$$

$$R^+(z) = \left[Q^+(z)+\sum_{n=0}^{N-1}q_n\left((1-\delta_p)\sum_{j=n}^{N-1}e_{j,n}\sum_{i=N-j}^{\infty}g_iz^{i+j}-z^n\right)\right]H(z), \tag{44}$$

where $M^{[k]}(z)=\sum_{n=0}^{\infty}m_n^{[k]}z^n=\tilde{\Psi}^{[k]}(\lambda-\lambda G(z))$ *and* $m_n^{[k]}=Pr\{n$ *customers arrive at the system during the period of* $k-th$ *type of vacation*$\}=\int_0^{\infty}\sum_{m=0}^{n}\frac{e^{-\lambda t}(\lambda t)^m}{m!}g_n^{(m)*}\psi^{[k]}(t)dt$, *where* $g_n^{(m)*}$ *is m-fold convolution of* g_n *with itself,* $H(z)=\sum_{n=0}^{\infty}h_nz^n=\tilde{R}(\lambda-\lambda G(z))$ *and* $h_n=Pr\{n$ *customers arrive during the set-up period*$\}=\int_0^{\infty}\sum_{m=0}^{n}\frac{e^{-\lambda t}(\lambda t)^m}{m!}g_n^{(m)*}r(t)dt$.

Proof. Multiplying (38)–(42) by corresponding power of z and y and summing them for $n=0$ to ∞ and k from 0 to $N-1$, we get

$$\left(\lambda-\theta-\lambda G(z)\right)\left(\sum_{k=0}^{a-1}\sum_{n=k}^{\infty}\bar{Q}_n^{[k]}(\theta)z^ny^k+\delta_p\sum_{k=a}^{N-1}\sum_{n=k}^{\infty}\bar{Q}_n^{[k]}(\theta)z^ny^k\right)=\sum_{n=0}^{a-1}\sum_{r=a}^{b}p_{n,r}(0)z^ny^n\tilde{\Psi}^{[n]}(\theta)$$
$$+\delta_p\sum_{n=0}^{N-1}q_n(0)z^ny^n\tilde{\Psi}^{[n]}(\theta)-\left(\sum_{k=0}^{a-1}\sum_{n=k}^{\infty}q_n^{[k]}(0)z^ny^k+\delta_p\sum_{k=a}^{N-1}\sum_{n=k}^{\infty}q_n^{[k]}(0)z^ny^k\right) \tag{45}$$

Now substituting $\theta=\lambda-\lambda G(z)$ in (45) and using (23), (25), (26), we get the required result (43).

Similarly, from (36)–(37), we obtain

$$\left(\lambda-\theta-\lambda G(z)\right)\sum_{n=N}^{\infty}\tilde{R}_n(\theta)z^n = \tilde{R}(\theta)\left[\sum_{n=0}^{\infty}q_n(0)z^n+\sum_{n=0}^{N-1}q_n(0)\left((1-\delta_p)\sum_{j=n}^{N-1}e_{j,n}\sum_{i=N-j}^{\infty}g_iz^{i+j}-z^n\right)\right]$$
$$-\sum_{n=N}^{\infty}r_n(0)z^n \tag{46}$$

Again substitution of $\theta=\lambda-\lambda G(z)$ in (46), use of (22), (24) and additional use of Lemma 4.1 leads the required result (44). □

Remark 5. The Eq. (43) represents the bivariate pgf of the number of customers waiting in the queue and the type of vacation availed by the server at the vacation completion epoch. By the appropriate inversion technique discussed later, one can achieve the corresponding probabilities. Similarly, Eq. (44) represents the single pgf of queue content at the end of set-up time and also we can calculate the corresponding probabilities using the appropriate inversion technique. Furthermore, Eqs. (43) and (44) will lead to a new additions in the existing literature as no such results are available so far to the best of the authors' knowledge.

Remark 6. If substitute $y=1$ in (43), we get the pgf of only queue content at the end of vacation time irrespective of the type of vacation and is given by

$$Q^+(z)=\sum_{n=0}^{a-1}p_n^+z^nM^{[n]}(z)+\delta_p\sum_{n=0}^{N-1}q_n^+z^nM^{[n]}(z) \tag{47}$$

Theorem 4.5. *The joint pgf of queue and server content at departure epoch is given by*

$$P^+(z,y) = \frac{\begin{array}{c}\sum_{n=0}^{a-1} p_n^+ z^n K^{(b)}(z) y^b \left(M^{[n]}(z)H(z) - 1 \right) \\ + \sum_{n=a}^{b-1} p_n^+ \left(z^b y^n K^{(n)}(z) - y^n K^{(b)}(z) K^{(n)}(z) + y^b K^{(b)}(z) K^{(n)}(z) - z^n y^b K^{(b)}(z) \right) \\ + \sum_{n=0}^{N-1} q_n^+ K^{(b)}(z) y^b H(z) \left((1-\delta_p) \sum_{j=n}^{N-1} e_{j,n} \sum_{i=N-j}^{\infty} g_i z^{i+j} + (\delta_p M^{[n]}(z) - 1) z^n \right)\end{array}}{z^b - K^{(b)}(z)} \tag{48}$$

where $K^{(r)}(z) = \tilde{S}_r(\lambda - \lambda G(z))$ *are the pgfs of* $k_n^{(r)}$ *and* $k_n^{(r)} = Pr\{n$ *arrivals during the service time of a batch size* $r\} = \int_0^\infty \sum_{m=0}^{n} \frac{e^{-\lambda t}(\lambda t)^m}{m!} g_n^{(m)*} s_r(t)dt$, $n \geq 0$ *and* $a \leq r \leq b$.

Proof. Multiplying (32)–(35) by corresponding powers of z and y and summing them for $n = 0$ to ∞ and r from a to b, we obtain

$$\begin{aligned}\left(\lambda - \theta - \lambda G(z)\right) \sum_{n=0}^{\infty} \sum_{r=a}^{b} \tilde{P}_{n,r}(z,y,\theta) = & \sum_{r=a}^{b} \sum_{m=a}^{b} p_{r,m}(0) y^r \tilde{S}_r(\theta) + \frac{\tilde{S}_b(\theta) y^b}{z^b} \Bigg[\sum_{n=0}^{\infty} \sum_{r=a}^{b} p_{n,r}(0) z^n + \sum_{n=N}^{\infty} r_n(0) \\ & z^n - \sum_{n=0}^{b} \sum_{r=a}^{b} p_{n,r}(0) z^n \Bigg] - \sum_{n=0}^{\infty} \sum_{r=a}^{b} p_{n,r}(0) z^n y^r\end{aligned} \tag{49}$$

Substituting $\theta = \lambda - \lambda G(z)$ in (49) and after using of Lemma 4.1, Eqs. (20), (21), (22) and (44), we get

$$\begin{aligned}P^+(z,y) = & \sum_{n=a}^{b} p_n^+ y^n K^{(b)}(z) + \frac{K^{(b)}(z) y^b}{z^b} \Bigg[P^+(z) + \Bigg(\sum_{n=0}^{a-1} p_n^+ z^n M^{[n]}(z) + \delta_p \sum_{n=0}^{N-1} q_n^+ z^n M^{[n]}(z) \\ & + \sum_{n=0}^{N-1} q_n^+ \left((1-\delta_p) \sum_{j=n}^{N-1} e_{j,n} \sum_{i=N-j}^{\infty} g_i z^{i+j} - z^n \right) \Bigg) H(z) - \sum_{n=0}^{b} p_n^+ z^n \Bigg]\end{aligned} \tag{50}$$

Substituting $y = 1$ in (50) and after some simple algebraic calculation, we get the pgf $P^+(z)$ of only queue content at departure epoch and is given by:

$$P^+(z) = \frac{\begin{array}{c}\sum_{n=0}^{a-1} p_n^+ z^n K^{(b)}(z) \left(M^{[n]}(z)H(z) - 1 \right) + \sum_{n=a}^{b-1} p_n^+ \left(K^{(n)}(z) z^b - K^{(b)}(z) z^n \right) + \\ \sum_{n=0}^{N-1} q_n^+ \left((1-\delta_p) \sum_{j=n}^{N-1} e_{j,n} \sum_{i=N-j}^{\infty} g_i z^{i+j} + (\delta_p M^{[n]}(z) - 1) z^n \right) H(z) K^{(b)}(z)\end{array}}{z^b - K^{(b)}(z)} \tag{51}$$

Finally, putting the expression of $P^+(z)$ in (50), we obtain our main result (48). □

Remark 7. Eq. (48) delineates the bivariate pgf of both queue length and server content distribution at the epoch of service completion, a pivotal aspect of this investigation. To the best of the authors' knowledge, such an outcome has not been previously reported in the existing literature. Consequently, it is hopeful that this result will foster a fresh advancement in the queueing literature. Furthermore, Eq. (51) represents the pgf of only queue length distribution at departure epoch which is also a new addition to the literature to the best of the authors' knowledge.

5. Computational procedure to determine the unknowns and the extraction of probabilities

In this section, our center of attention is to extract the probabilities $p_{n,r}^+$, $n \geq 0$, $a \leq r \leq b$, $q_n^{[k]+}$, $n \geq k$, $0 \leq k \leq a-1$ or $0 \leq k \leq N-1$ and r_n^+, $n \geq 0$ from (48), (43) and (44), respectively. Before extracting the probabilities, we need to find out the unknown probabilities p_n^+, $n = 0, \ldots, b-1$ and q_n^+, $n = 0, \ldots, N-1$ which are appearing in the numerator of (48), (43) and (44). In order to calculate the value of those probabilities, we follow the same procedure given in Pradhan and Gupta [6]. It is clear that the numerator of (48) contains the total $b+N$ unknowns. We know that the pgf is analytic within the unit disk $|z| \leq 1$. This implies that the roots with absolute value less than 1 of the denominator of (48) must be the roots of the numerator of (48). Using Rouche's theorem, we know that $z^b - K^{(b)}(z) = 0$ has exactly b roots (say, $\alpha_1, \alpha_2, \ldots, \alpha_b = 1$) in the unit disk $|z| \leq 1$. Substituting those, we get $b-1$ equations and after using L'Hospital's rule for the root $\alpha_b = 1$, we obtain another equation. We use the result given below in Lemma 5.1. By which, we can express q_n^+, $n = 0, \ldots, N-1$ in terms of p_n^+, $n = 0, \ldots, a-1$. This reduces the total unknowns from $b+N$ to b. Now solving 'b' linear equations using suitable numerical methods, we get the required 'b' unknowns.

Lemma 5.1. *The relation between* p_n^+ *and* q_n^+ *is given by:*

$$q_n^+ = \sum_{j=0}^{n} \xi_{n-j}^{[j]} p_j^+, \qquad 0 \leq n \leq a-1, \tag{52}$$

$$q_a^+ = \frac{\pi_a}{1-\delta_p m_0^{[a]}}, \quad q_n^+ = \frac{\pi_n + \delta_p \sum_{j=a}^{n-1} m_{n-j}^{[j]} q_j^+}{1-\delta_p m_0^{[n]}}, \qquad a+1 \le n \le N-1, \tag{53}$$

where $\xi_0^{[j]} = \dfrac{m_0^{[j]}}{1-\delta_p m_0^{[j]}}$ *and* $\xi_n^{[j]} = \dfrac{m_n^{[j]} + \delta_p \sum_{i=1}^{n} m_i^{[n+j-i]} \xi_{n-i}^{[j]}}{1-\delta_p m_0^{[n+j]}}, \quad 0 \le j \le a-1, \quad 1 \le n \le a-1,$ *such that* $n+j \le a-1$ *and* $\pi_n = \sum_{i=0}^{a-1}\left(m_{n-i}^{[i]} + \delta_p \sum_{j=0}^{a-1-i} \xi_j^{[i]} m_{n-i-j}^{[i+j]}\right) p_i^+, \quad a \le n \le N-1.$

The use of these above expressions mainly reduce the number of unknown probabilities appearing in the bivariate pgf (48). Here, the main objective is to express q_n^+ $(n = 0, 1, \dots, N-1)$ in terms of p_n^+ $(n = 0, 1, \dots, a-1)$ in order to simplify the computational process.

Proof. Eq. (47) can be written in the following form:

$$\begin{aligned}
\sum_{n=0}^{\infty} q_n^+ z^n &= \sum_{n=0}^{a-1} (p_n^+ + \delta_p q_n^+) z^n M^{[n]}(z) + \delta_p \sum_{n=a}^{N-1} q_n^+ z^n M^{[n]}(z) \\
&= \sum_{n=0}^{a-1} (p_n^+ + \delta_p q_n^+) z^n \sum_{j=0}^{\infty} m_j^{[n]} z^j + \delta_p \sum_{n=a}^{N-1} q_n^+ z^n \sum_{j=0}^{\infty} m_j^{[n]} z^j \\
&= \sum_{n=0}^{a-1} \left[\sum_{j=0}^{n} m_{n-j}^{[j]} (p_j^+ + \delta_p q_j^+)\right] z^n + \sum_{n=a}^{\infty} \left[\sum_{j=0}^{a-1} m_{n-j}^{[j]} (p_j^+ + \delta_p q_j^+)\right] z^n \\
&\quad \delta_p \sum_{n=a}^{N-1} \left[\sum_{j=a}^{n} m_{n-j}^{[j]} q_j^+\right] z^n + \delta_p \sum_{n=N}^{\infty} \left[\sum_{j=a}^{N-1} m_{n-j}^{[j]} q_j^+\right] z^n
\end{aligned} \tag{54}$$

Now equating the coefficient of z^n, $n = 0, 1, \dots, a-1$ from both the sides of (54), we acquire for $n = 0$

$$q_0^+ = \frac{m_0^{[0]}}{1-\delta_p m_0^{[0]}} p_0^+ = \xi_0^{[0]} p_0^+, \qquad \text{where } \xi_0^{[0]} = \frac{m_0^{[0]}}{1-\delta_p m_0^{[0]}}$$

For $n = 1$, we get

$$q_1^+ = \xi_1^{[0]} p_0^+ + \xi_0^{[1]} p_1^+ = \sum_{j=0}^{1} \xi_{1-j}^{[j]} p_j^+, \qquad \text{where } \xi_1^{[0]} = \frac{m_1^{[0]} + \delta_p m_1^{[0]} \xi_0^{[0]}}{1-\delta_p m_0^{[1]}}, \quad \xi_0^{[1]} = \frac{m_0^{[1]}}{1-\delta_p m_0^{[1]}}.$$

In the similar way, we obtain the required results for other values of n.

Further equating the coefficient of z^n, $n = a, a+1, \dots, N-1$ from both the sides of (54), for $n = a$, we get

$$\begin{aligned}
q_a^+ &= \sum_{j=0}^{a-1} m_{a-j}^{[j]} \left(p_j^+ + \delta_p q_j^+\right) + \delta_p m_0^{[a]} q_a^+ \\
\implies \left(1-\delta_p m_0^{[a]}\right) q_a^+ &= \sum_{i=0}^{a-1} \left[m_{a-i}^{[i]} + \delta_p \sum_{j=0}^{a-1-i} \xi_j^{[i]} m_{a-i-j}^{[i+j]}\right] p_i^+ \\
\implies q_a^+ &= \frac{\pi_a}{1-\delta_p m_0^{[a]}}.
\end{aligned}$$

where $\pi_a = \sum_{i=0}^{a-1} \left[m_{a-i}^{[i]} + \delta_p \sum_{j=0}^{a-1-i} \xi_j^{[i]} m_{a-i-j}^{[i+j]}\right] p_i^+$

Also for $n = a+1, a+2, \dots, N-1$, we get

$$\begin{aligned}
q_n^+ &= \sum_{j=0}^{a-1} m_{n-j}^{[j]} \left(p_j^+ + \delta_p q_j^+\right) + \sum_{j=a}^{n} m_{n-j}^{[j]} q_j^+ \\
\implies \left(1-\delta_p m_0^{[n]}\right) q_n^+ &= \sum_{i=0}^{a-1} \left[m_{n-i}^{[i]} + \delta_p \sum_{j=0}^{a-1-i} \xi_j^{[i]} m_{n-i-j}^{[i+j]}\right] p_i^+ + \delta_p \sum_{j=a}^{n-1} m_{n-j}^{[j]} q_j^+ \\
\implies q_n^+ &= \frac{\pi_n + \delta_p \sum_{j=a}^{n-1} m_{n-j}^{[j]} q_j^+}{1-\delta_p m_0^{[n]}}
\end{aligned}$$

where $\pi_n = \sum_{i=0}^{a-1} \left[m_{n-i}^{[i]} + \delta_p \sum_{j=0}^{a-1-i} \xi_j^{[i]} m_{n-i-j}^{[i+j]}\right] p_i^+, \quad n = a+1, a+2, \dots, N-1.$ □

5.1. Extraction of joint probabilities from the known bivariate pgf

After determining all the unknowns $\{p_n^+\}_{n=0}^{b-1}$ and $\{q_n^+\}_{n=0}^{N-1}$, the pgfs are completely known to us. Now we focus on the extraction of the joint probabilities $p_{n,r}^+$, $n \ge 0$, $a \le r \le b$, the queue length distribution $q_n^{[k]+}$, $n \ge k$, $0 \le k \le a-1$ or $0 \le k \le N-1$, r_n^+, $n \ge 0$

at the end of kth type vacation, set-up time, respectively. After the extraction, one can easily derive: (i) the joint probabilities at arbitrary/random epoch, (ii) several marginal distributions that are required for computing the performance measures. This can be achieved by the inversion of $P^{+}(z,y)$, which is not always easily tractable because of the complexity in the expression. Firstly, we assemble the coefficients of $y^r,\ a \le r \le b$ from both sides of (48).

Coefficients of $y^r,\ a \le r \le b-1$:

$$\sum_{n=0}^{\infty} p_{n,r}^{+} z^n = p_r^{+} K^{(r)}(z) \tag{55}$$

Coefficient of y^b:

$$\sum_{n=0}^{\infty} p_{n,b}^{+} z^n = \frac{\sum_{n=0}^{a-1} p_n^{+} z^n K^{(b)}(z)\Big(M^{[n]}(z)H(z)-1\Big) + \sum_{n=a}^{b-1} p_n^{+}\Big(K^{(b)}(z)K^{(n)}(z) - z^n K^{(b)}(z)\Big) + \sum_{n=0}^{N-1} q_n^{+} K^{(b)}(z)H(z)\Big((1-\delta_p)\sum_{j=n}^{N-1} e_{j,n}\sum_{i=N-j}^{\infty} g_i z^{i+j} + (\delta_p M^{[n]}(z)-1)z^n\Big)}{z^b - K^{(b)}(z)} \tag{56}$$

Collecting the coefficients of z^n from both sides of (55), we obtain

$$p_{n,j}^{+} = p_j^{+} k_n^{(j)}, \quad n \ge 0, \quad a \le j \le b-1 \tag{57}$$

Now we need to calculate $p_{n,b}^{+}$ from (56), for which the inversion of the right-hand side of (56) is required. However it is not an easy task due to the complex expression. For known service time, vacation time and set-up time distributions, the right side of (56) is a completely known function of z. After substituting $K^{(r)}(z) = \tilde{S}_r(\lambda - \lambda G(z)),\ a \le r \le b$, $M^{[n]}(z) = \tilde{\Psi}^{[n]}(\lambda - \lambda G(z))$ and $H(z) = \tilde{R}(\lambda - \lambda G(z))$ in (56), let us denote numerator of (56) as $N_1(z)$ and denominator as $N_2(z)$ with degrees ℓ_1 and ℓ_2, respectively. Now, we obtain $p_{n,b}^{+}$ in terms of the roots of $N_2(z) = 0$ by discussing the following cases.

- **When all the zeros of $N_2(z)$ in $|z| > 1$ are distinct**
  Let us assume $\gamma_1, \gamma_2, \ldots, \gamma_{\ell_2}$ to be the distinct roots of $N_2(z) = 0$, out of which $\ell_2 - b$ are in $|z| > 1$.

Case 1: $\ell_1 \ge \ell_2$

Applying the partial-fraction expansion, $\sum_{n=0}^{\infty} p_{n,b}^{+} z^n$ can be uniquely written as:

$$\sum_{n=0}^{\infty} p_{n,b}^{+} z^n = \sum_{i=0}^{\ell_1-\ell_2} \tau_i z^i + \sum_{k=1}^{\ell_2-b} \frac{c_k}{\gamma_k - z} \tag{58}$$

for some τ_i and c_k's, with τ_i as the resultant coefficient after division of $N_1(z)$ by $N_2(z)$ and $c_k = -\frac{N_1(\gamma_k)}{N_2'(\gamma_k)}, \quad k = 1, 2, \ldots, \ell_2 - b$.

Now, collecting the coefficient of z^n from both the sides of (58), we get

$$p_{n,b}^{+} = \begin{cases} \tau_n + \displaystyle\sum_{k=1}^{\ell_2-b} \frac{c_k}{\gamma_k^{n+1}}, & 0 \le n \le \ell_1 - \ell_2, \\ \displaystyle\sum_{k=1}^{\ell_2-b} \frac{c_k}{\gamma_k^{n+1}}, & n > \ell_1 - \ell_2. \end{cases} \tag{59}$$

Case 2: $\ell_1 < \ell_2$

The first term does not appear in (59) and the remaining term provides the joint distribution and $p_{n,b}^{+},\ \ n \ge 0$ is given by:

$$p_{n,b}^{+} = \sum_{k=1}^{\ell_2-b} \frac{c_k}{\gamma_k^{n+1}}, \quad n \ge 0. \tag{60}$$

Remark 8. Whenever $N_2(z)$=0 has multiple roots in $|z| > 1$, with the slight modification of the above partial fraction procedure, we can acquire the joint distribution, which is thoroughly discussed by Pradhan [33].

6. The joint distribution of queue and server content at arbitrary epoch

In order to calculate the several marginal distributions (which will be used for computing system performances), the system designer requires the queue and server content distribution at an arbitrary epoch. Obtaining joint distribution at arbitrary epoch from (32)–(42), using the concept of pgf is a tedious task. To overcome this difficulty, we generate the following relationship with departure epoch probabilities by which we can easily calculate the joint distribution at arbitrary epochs after deriving the joint distribution at the departure epoch.

Theorem 6.1. *The probabilities $\{p_{n,r}^{+},\ q_n^{[k]+},\ r_n^{+}\}$ and $\{p_{n,r},\ q_n^{[k]},\ r_n\}$ are related by*

$$v_n = \frac{(1-\delta_p)\sum_{m=0}^{n} e_{n,m} q_m^{+}}{E^*}, \quad 0 \le n \le N-1. \tag{61}$$

$$p_{0,r} = \frac{1}{E^*}(p_r^+ - p_{0,r}^+), \quad a \le r \le b. \tag{62}$$

$$p_{n,r} = \sum_{j=1}^{n} g_j p_{n-j,r} - \frac{1}{E^*} p_{n,r}^+ \quad n \ge 1, \quad a \le r \le b-1. \tag{63}$$

$$p_{n,b} = \sum_{j=1}^{n} g_j p_{n-j,b} + \frac{1}{E^*}(p_{n+b}^+ - p_{n,b}^+), \quad 1 \le n \le N-b-1. \tag{64}$$

$$p_{n,b} = \sum_{j=1}^{n} g_j p_{n-j,b} + \frac{1}{E^*}(p_{n+b}^+ - p_{n,b}^+ + r_{n+b}^+), \quad n \ge N-b. \tag{65}$$

$$q_0^{[0]} = \frac{1}{E^*}\left(p_0^+ - (1-\delta_p) q_0^{[0]+}\right) \tag{66}$$

$$q_k^{[k]} = \frac{1}{E^*}(p_k^+ - q_k^{[k]+} + \delta_p q_k^+), \quad 0 \le k \le a-1. \tag{67}$$

$$q_k^{[k]} = \frac{1}{E^*}(q_k^+ - q_k^{[k]+}), \quad a \le k \le N-1. \tag{68}$$

$$q_n^{[k]} = \sum_{j=0}^{n} g_j q_{n-j}^{[k]} - \frac{1}{E^*} q_n^{[k]+}, \quad 0 \le k \le a-1, \quad n \ge k+1. \tag{69}$$

$$q_n^{[k]} = \sum_{j=0}^{n} g_j q_{n-j}^{[k]} - \frac{1}{E^*} q_n^{[k]+}, \quad a \le k \le N-1, \quad n \ge k+1. \tag{70}$$

$$r_N = \frac{1}{E^*}\left(q_N^+ + (1-\delta_p)\sum_{j=0}^{N-1}\sum_{m=0}^{j} e_{j,m} g_{N-j} q_m^+ - r_N^+\right) \tag{71}$$

$$r_n = \sum_{j=N}^{n-1} g_{n-j} r_j + \frac{1}{E^*}\left(q_n^+ + (1-\delta_p)\sum_{j=0}^{N-1}\sum_{m=0}^{j} e_{j,m} g_{N-j} q_m^+ - r_n^+\right), \quad n \ge N+1 \tag{72}$$

where $E^* = \lambda\omega + (1-\delta_p)\sum_{j=0}^{N-1}\sum_{m=j}^{N-1} e_{m,j} q_j^+$ *and* ω *is given in Lemma 4.3.*

Proof. In case of multiple vacations $\delta_p = 1$, hence Eq. (61) does not appear. For single vacation, putting $\delta_p = 0$ in (1) and dividing by $\Omega = \sum_{n=0}^{\infty}\sum_{r=a}^{b} p_{n,r}(0) + (1-\delta_p)\sum_{n=0}^{\infty}\sum_{k=0}^{min(n,a-1)} q_n^{[k]}(0) + \delta_p \sum_{n=0}^{\infty}\sum_{k=0}^{min(n,N-1)} q_n^{[k]}(0) + \sum_{n=N}^{\infty} r_n(0)$, using (26) and (31), we obtain

$$\lambda\omega v_0 = \left(1 - (1-\delta_p)\sum_{n=0}^{N-1} v_n\right) q_0^+. \tag{73}$$

Similarly, from (28), we get

$$\lambda\omega v_n = \left(1 - (1-\delta_p)\sum_{n=0}^{N-1} v_n\right)\sum_{m=0}^{n} e_{n,m} q_m^+, \quad 1 \le n \le N-1. \tag{74}$$

Dividing (74) by (73) and after some algebraic manipulation, we acquire

$$\sum_{n=0}^{N-1} v_n = \frac{v_0}{q_0^+}\sum_{j=0}^{N-1}\sum_{m=j}^{N-1} e_{m,j} q_j^+. \tag{75}$$

Using (75) in (73), we obtain

$$v_0 = \frac{q_0^+}{E^*}, \tag{76}$$

where $E^* = \lambda\omega + (1-\delta_p)\sum_{j=0}^{N-1}\sum_{m=j}^{N-1} e_{m,j} q_j^+$. Using (76) in (73), we get

$$1 - (1-\delta_p)\sum_{n=0}^{N-1} v_n = \frac{\lambda\omega}{E^*} \tag{77}$$

using (77) in (74), we get

$$v_n = \frac{\sum_{m=0}^{n} e_{n,m} q_m^+}{E^*}, \qquad 1 \le n \le N-1, \tag{78}$$

which is the desired result given in (61).

Now, setting $\theta = 0$ in (32)–(35), we get

$$\lambda p_{0,r} = \sum_{m=a}^{b} p_{r,m}(0) - p_{0,r}(0), \quad a \le r \le b, \tag{79}$$

$$\lambda p_{n,r} = \lambda \sum_{j=1}^{n} g_j p_{n-j,r} - p_{n,r}(0), \quad a \le r \le b-1, \quad n \ge 1, \tag{80}$$

$$\lambda p_{n,b} = \lambda \sum_{j=1}^{n} g_j p_{n-j,b} + \sum_{r=a}^{b} p_{n+b,r}(0) - p_{n,b}(0), \quad 1 \le n \le N-b-1, \tag{81}$$

$$\lambda p_{n,b} = \lambda \sum_{j=1}^{n} g_j p_{n-j,b} + \sum_{r=a}^{b} p_{n+b,r}(0) + r_{n+b}(0) - p_{n,b}(0), \quad n \ge N-b. \tag{82}$$

Dividing (79) by $\sum_{n=0}^{\infty}\sum_{r=a}^{b} p_{n,r}(0) + \left(1-\delta_p\right)\sum_{n=0}^{\infty}\sum_{k=0}^{\min(n,a-1)} q_n^{[k]}(0) + \delta_p \sum_{n=0}^{\infty}\sum_{k=0}^{\min(n,N-1)} q_n^{[k]}(0) + \sum_{n=N}^{\infty} r_n(0)$ and using (15), (25), (31), and (77), we obtain

$$p_{0,r} = \frac{p_r^+ - p_{0,r}^+}{E^*}, \quad a \le r \le b. \tag{83}$$

which is the desired result in (62). Now dividing (80) by $\sum_{n=0}^{\infty}\sum_{r=a}^{b} p_{n,r}(0) + (1-\ \delta_p)\sum_{n=0}^{\infty}\sum_{k=0}^{\min(n,a-1)}$ $q_n^{[k]}(0) + \delta_p \sum_{n=0}^{\infty}\sum_{k=0}^{\min(n,N-1)} q_n^{[k]}(0) + \sum_{n=N}^{\infty} r_n(0)$ and using (25), (31), and (77), we obtain

$$p_{n,r} = \sum_{j=1}^{n} g_j p_{n-j,r} - \frac{1}{E^*} p_{n,r}^+, \quad a \le r \le b-1. \tag{84}$$

which is desired result in (63). Proceeding in the similar fashion, we obtain (64)–(65) as our desired results. However, substituting $\theta = 0$ in (36)–(42) and then dividing by $\sum_{n=0}^{\infty}\sum_{r=a}^{b} p_{n,r}(0) + \left(1-\delta_p\right)$ $\sum_{n=0}^{\infty}\sum_{k=0}^{\min(n,a-1)} q_n^{[k]}(0) + \delta_p \sum_{n=0}^{\infty}\sum_{k=0}^{\min(n,N-1)} q_n^{[k]}(0) + \sum_{n=N}^{\infty} r_n(0)$, using the similar approach, we acquire the required results (66)–(72). □

Remark 9. In a group-arrival queueing system, joint probabilities at pre-arrival epoch differ from the arbitrary epoch probabilities. Here, those can be easily found using the concept given by Pradhan and Gupta [6] [Page 25, Section 5]. Although we have computed pre-arrival epoch probabilities numerically, we have not displayed due to lack of space.

7. Marginal distributions and performance measures

In this section, we provide some marginal distributions and significant performance measures from practical and designing point of view.

- The distribution of system content (which includes the total number of customers in the queue as well as with server), p_n^{sys} $(n \ge 0)$, is given by

$$p_n^{sys} = \begin{cases} (1-\delta_p)\upsilon_n + \sum_{k=0}^{min(a-1,n)} q_n^{[k]}, & 0 \le n \le a-1, \\ (1-\delta_p)\upsilon_n + \sum_{k=0}^{min(a-1,n)} q_n^{[k]} + \delta_p \sum_{k=a}^{min(N-1,n)} q_n^{[k]} + \sum_{r=a}^{min(b,n)} p_{n-r,r}, & a \le n \le N-1, \\ \sum_{r=a}^{b} p_{n-r,r} + \sum_{k=0}^{min(a-1,n)} q_n^{[k]} + \delta_p \sum_{k=a}^{min(N-1,n)} q_n^{[k]} + r_n, & n \ge N. \end{cases}$$

- The distribution of the number of customers in the queue, p_n^{queue} $(n \ge 0)$, is given by,

$$p_n^{queue} = \begin{cases} (1-\delta_p)\upsilon_n + \sum_{r=a}^{b} p_{n,r} + \sum_{k=0}^{min(a-1,n)} q_n^{[k]} + \delta_p \sum_{k=a}^{min(N-1,n)} q_n^{[k]}, & 0 \le n \le N-1, \\ \sum_{r=a}^{b} p_{n,r} + \sum_{k=0}^{min(a-1,n)} q_n^{[k]} + \delta_p \sum_{k=a}^{min(N-1,n)} q_n^{[k]} + r_n, & n \ge N. \end{cases}$$

- The probability that the server is busy with some customers (P_{busy}), in the vacation state (Q_{vac}), and in the dormant state (P_{dor}), in set-up state (P_{set}), in idle state (P_{idle}) are given by $P_{busy} = \sum_{n=0}^{\infty}\sum_{r=a}^{b} p_{n,r}$, $Q_{vac} = \sum_{n=0}^{\infty}\sum_{k=0}^{min(a-1,n)} q_n^{[k]} + \delta_p \sum_{n=0}^{\infty}\sum_{k=a}^{min(N-1,n)} q_n^{[k]}$, $P_{dor} = \sum_{n=0}^{N-1} \upsilon_n$, $P_{set} = \sum_{n=N}^{\infty} r_n$, $P_{idle} = (1-\delta_p)P_{dor} + Q_{vac} + P_{set}$, respectively.
- The probability that r customers with the server at arbitrary epoch, p_r^{ser} $(a \le r \le b)$, is given by $p_r^{ser} = \sum_{n=0}^{\infty} p_{n,r}$.

- The expected number of customers in the queue at arbitrary time $(L_q) = \sum_{n=0}^{\infty} n p_n^{queue}$.
- The expected number of the customers in the system at arbitrary time $(L) = \sum_{n=0}^{\infty} n p_n^{sys}$.
- The expected number of customers with the server $(L_s) = \sum_{r=a}^{b} r p_r^{ser}$.
- The expected waiting time of a customer in the queue $(W_q) = \dfrac{L_q}{\lambda \bar{g}}$ as well as in system $(W) = \dfrac{L}{\lambda \bar{g}}$.

7.1. Expected length of idle and busy period

In this subsection, first, we concentrate on the derivation of the mean (expected) length of idle and busy period for the concerned model which are pivotal performance measures. These quantities contribute significantly for constructing the cost function of the system per unit of time. In the concerned model, the idle period consists of the dormant, vacation and set-up period in case of a single vacation. Whereas in the case of multiple vacations, the idle period is the sum of one or more vacation period(s) and set-up period only. In case of a single vacation, the busy period is the time interval during which the server provides service continuously.

It is terminated when the server finds that the number of waiting customers is less than 'a' after completing the service of a batch. The vacation period is initiated when the number of customers at departure epoch is less than 'a' and it depends on the number of customers. The dormant period is started when the number of waiting customers in the queue at the end of vacation is less than 'N' and is terminated when the number of aggregated customers reaches the threshold value 'N'. At the end of the dormant period, the set-up period starts. After the completion of set-up, the server commences the service process. In case of multiple vacations, the vacation period is initiated when the server finds that the waiting customers in the queue are less than 'a' after the completion of the service of a batch and is terminated when the queue size reaches the threshold value 'N' at the end of a vacation. In this case, the vacation period may be constituted by one or more vacation period(s). Since the concerned model consists of both single and multiple vacations, we derive the mean length of idle and busy period by considering the two cases.

Case 1: The system operates under a single vacation policy: Let I be the RV corresponding to the idle period. As we know the expected length of the idle period is the sum of the expected length of dormant period, the expected length of the vacation period and the expected length of the set-up time and hence $E[I] = E[D] + E[I_1] + E[R]$ where D is the RV for the dormant period, I_1 is the RV for the vacation period of the single vacation policy and R is the RV corresponding to set-up time.

Expected length of dormant period: We already introduced that $\widehat{N}_q(t)$ denotes the number of waiting customers in the queue at time t. At the end of a busy period, if the queue size is less than 'a', then $\widehat{N}_q(t)$ moves into one of the vacation states $\Lambda = \{0, 1, \dots, a-1\}$. Now the conditional probability that $\widehat{N}_q(t)$ enters into 'i' ($\in \Lambda$) state at the end of a service period is given by $\dfrac{p_i^+}{\sum_{i=0}^{a-1} p_i^+}$ and the server enters into 'i'-th type vacation. For fixed $i \in \Lambda$, suppose j ($\leq N-1-i$) is the number of customers arrive during the ith type vacation period with probability $m_j^{[i]}$. Then the server moves into the dormant states $\Gamma = \{\zeta : \zeta = i + j,\ 0 \leq i \leq a-1 \text{ and } 0 \leq j \leq N-1-i\}$ and leaves the dormant state when at least $N-i-j$ customers arriving at the system. The customers arrive in batches with mean batch size $\bar{g}$, and the inter-arrival time between two consecutive batch arrivals are i.i.d RVs and also exponentially distributed with mean $\dfrac{1}{\lambda}$. So the expected time that $N-i-j$ customers arrive is $\dfrac{N-i-j}{\lambda \bar{g}}$. By considering all the possible values of i, we get

$$E[D] = \frac{1}{\sum_{i=0}^{a-1} p_i^+} \left[\sum_{i=0}^{a-1} p_i^+ \sum_{j=0}^{N-1-i} m_j^{[i]} \left(\frac{N-i-j}{\lambda \bar{g}} \right) \right]$$

Expected length of vacation period: We know that the vacation period starts at the end of a service period if the queue size less than 'a' and suppose there are 'k' ($0 \leq k \leq a-1$) customers with probability $\dfrac{p_k^+}{\sum_{i=0}^{a-1} p_i^+}$, then the server goes for kth vacation and the duration of the vacation period is $\bar{\psi}^{[k]}$. Now considering all the possible values of k, we can easily get $E[I_1] = \dfrac{\sum_{k=0}^{a-1} p_k^+ \bar{\psi}^{[k]}}{\sum_{i=0}^{a-1} p_i^+}$.

Now the expected length of the idle period is given by

$$\begin{aligned} E[I] &= E[D] + E[I_1] + E[R] \\ \implies E[I] &= \frac{1}{\sum_{i=0}^{a-1} p_i^+} \left[\sum_{i=0}^{a-1} p_i^+ \sum_{j=0}^{N-1-i} m_j^{[i]} \left(\frac{N-i-j}{\lambda \bar{g}} \right) \right] + \frac{\sum_{k=0}^{a-1} p_k^+ \bar{\psi}^{[k]}}{\sum_{i=0}^{a-1} p_i^+} + \bar{r} \end{aligned}$$

where $E[R] = \bar{r}$ which is already mentioned in the model description.

Expected length of cycle: The length of a cycle is the time from the beginning of the last busy period to the beginning of the next busy period. More precisely, the cycle length consists of the duration of one or more consecutive services (without any break) and idle duration (whenever the minimum threshold value is not accumulated). The idle period consists of only one vacation period, a dormant period and a set-up time. The starting points of two consecutive busy periods are considered as renewal points. The cycle length along with renewal points are clearly exhibited in Fig. 3 for a single vacation. Mutual independence of arrival process, service process, vacation period, and set-up process lead to the length of the cycle renewal points as sequence of i.i.d RVs. This eventually leads to no correlation among cycle lengths. The length of cycle of the concerned model is denoted by L_c. Therefore, L_c is the sum of vacation period, dormant period, set-up period and busy period which implies $E[L_c] = E[I_1] + E[D] + E[R] + E[B]$, where B is RV corresponding to busy period.

Expected length of busy period: We already know that $P_{idle} = P_{dor} + Q_{vac} + P_{set}$ and from the theory of alternating renewal process (see Wang et al. [34], page 2204), we get $P_{dor} = \dfrac{E[D]}{E[L_c]}$, $Q_{vac} = \dfrac{E[I_1]}{E[L_c]}$ and $P_{set} = \dfrac{E[R]}{E[L_c]}$ and hence $P_{idle} = \dfrac{E[D]}{E[L_c]} + \dfrac{E[I_1]}{E[L_c]} + \dfrac{E[R]}{E[L_c]} = \dfrac{E[I]}{E[L_c]}$. In a similar way, we can write that

$$\begin{aligned} \frac{E[B]}{E[I]} &= \frac{\dfrac{E[B]}{E[L_c]}}{\dfrac{E[I]}{E[L_c]}} = \frac{\dfrac{E[L_c] - E[I]}{E[L_c]}}{\dfrac{E[I]}{E[L_c]}} = \frac{1 - P_{idle}}{P_{idle}} \\ \implies E[B] &= E[I] \left(\frac{1 - P_{idle}}{P_{idle}} \right) \end{aligned}$$

Case 2: The system operates under multiple vacations policy: Let I be the RV corresponding to the idle period. Then the expected length of the idle period is the sum of the expected length of the vacation period due to multiple vacations and the expected length of set-up time, i.e., $E[I] = E[I_2] + E[R]$, where I_2 is the RV for multiple vacation periods.

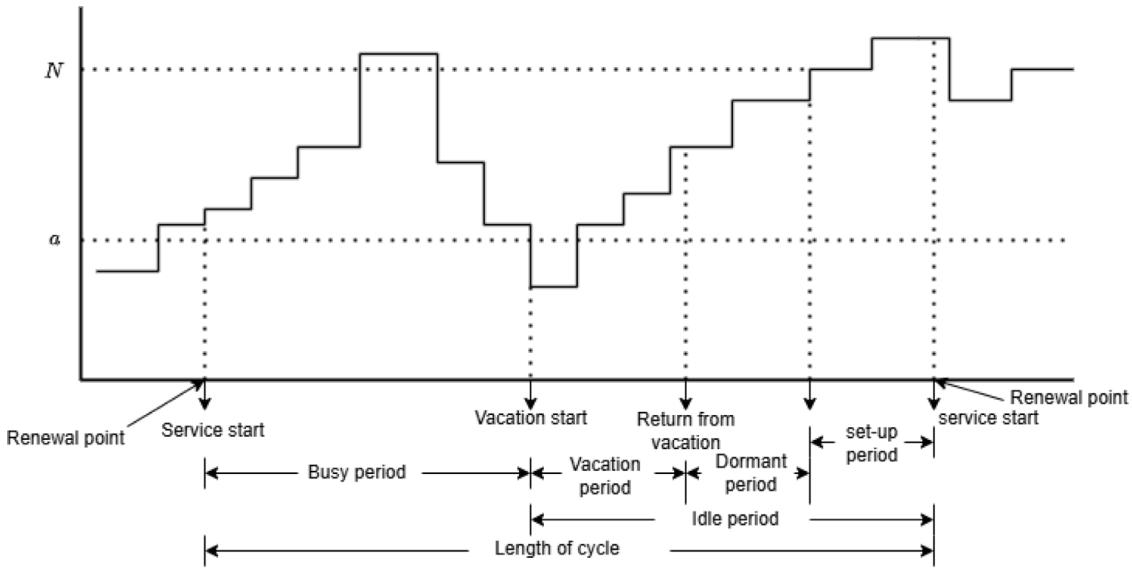

Fig. 3. A typical sample path of the concerned model for single vacation.

Expected length of vacation period: In case of multiple vacations, the vacation period may be constituted by one or more vacation period(s) which completely depend on the queue size at the end of vacation period. We define a new RV, U as follows:

$$U = \begin{cases} 0, & \text{if at least } N \text{ customers are available at the end of the first vacation,} \\ 1, & \text{if fewer than } N \text{ customers are available at the end of the first vacation.} \end{cases}$$

Now, using the total probability concept, we obtain

$$\begin{aligned} E[I_2] &= E[I_2|U=0]Pr\{U=0\} + E[I_2|U=1]Pr\{U=1\} \\ \implies E[I_2] &= E[I_1]Pr\{U=0\} + (E[I_1]+E[I_2])Pr\{U=1\} \\ \implies E[I_2] &= \frac{E[I_1]}{1-Pr\{U=1\}} = \frac{\dfrac{\sum_{k=0}^{a-1} p_k^+ \bar{\psi}^{[k]}}{\sum_{i=0}^{a-1} p_i^+}}{1-\sum_{n=0}^{a-1}\sum_{i=0}^{n} p_i^+ m_{n-i}^{[i]} - \sum_{n=a}^{N-1}\sum_{i=0}^{a-1} p_i^+ m_{n-i}^{[i]}} \end{aligned}$$

Now the expected length of the idle period is given by

$$\begin{aligned} E[I] &= E[I_2] + E[R] \\ \implies E[I] &= \frac{\dfrac{\sum_{k=0}^{a-1} p_k^+ \bar{\psi}^{[k]}}{\sum_{i=0}^{a-1} p_i^+}}{1-\sum_{n=0}^{a-1}\sum_{i=0}^{n} p_i^+ m_{n-i}^{[i]} - \sum_{n=a}^{N-1}\sum_{i=0}^{a-1} p_i^+ m_{n-i}^{[i]}} + \bar{r} \end{aligned}$$

Expected length of a cycle: Similar as the single vacation, in the case of multiple vacations, the length of a cycle is the time from the beginning of the last busy period to the beginning of the next busy period. Hence the length of a cycle is the sum of the idle period and busy period, where the idle period is the sum of one or more vacation period(s) and set-up time. The cycle length and renewal points are shown in Fig. 4, in cases of Multiple vacations. Therefore, the expected length of the cycle is given by $E[L_c] = E[I]+E[B]$. Since the various stochastic processes such as arrival process, service process, vacation periods, and set-up time are mutually independent, we can conclude that the length of cycles forms a sequence of i.i.d RVs which implies that no correlation is there among the cycles.

Expected length of busy period: Using the theory of alternative renewal process, the expected length of a busy period is given by

$$\begin{aligned} \frac{E[B]}{E[I]} &= \frac{\dfrac{E[B]}{E[L_c]}}{\dfrac{E[I]}{E[L_c]}} = \frac{\dfrac{E[L_c]-E[I]}{E[L_c]}}{\dfrac{E[I]}{E[L_c]}} = \frac{1-P_{idle}}{P_{idle}} \\ \implies E[B] &= E[I]\left(\frac{1-P_{idle}}{P_{idle}}\right) \end{aligned}$$

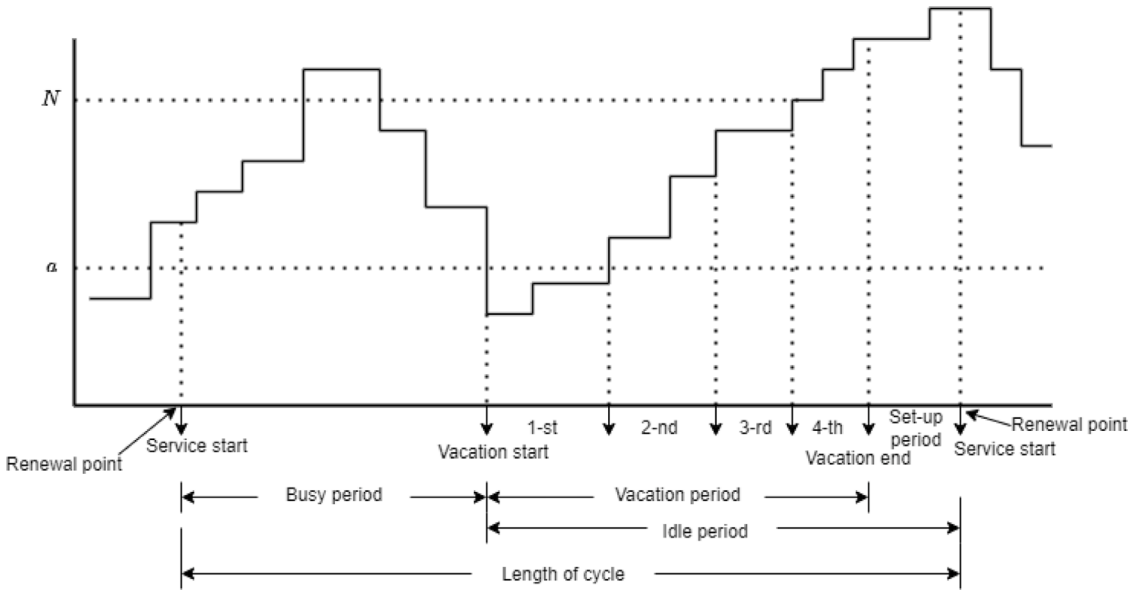

Fig. 4. A typical sample path of the concerned model for multiple vacation.

8. Numerical illustration

In this section, our main motive is to demonstrate the numerical results by which our proposed methodology and the theoretical results (derived in the previous sections) may be justified. It also explores the effect of parameters on the system performance by considering some important probability distributions as service, vacation and set-up time. We also develop a cost function and investigate the effect of parameters on total system cost. For different service and vacation time distributions, the impact of the system's parameters have also been studied. We have performed extensive experiments using Maple 16 on PC having configuration Intel(R) Core(TM) i5-6200U CPU Processor @ 2.30 GHz 2.40 GHz with 8.00 GB of RAM, but due to lack of space only a few of them are presented here in the form of tables and graphs which provide the inner feeling to the readers as well as practitioners. Here it should be mentioned that in real-life scenarios, the size of arrival groups of customers is finite. Keeping this in mind, we consider the finite size of arrival batches during the computational process.

Example: WSNs are an integral part of the modern telecommunication system, and they consists of numerous sensor nodes. The main task of the sensor nodes is to collect the data and transmit it to the central location i.e., the base station. In practice, it may be found that several sensor nodes are kept outside the communication range of the base station. In such scenarios, other nodes, generally known as relay nodes are positioned in between with the aim of collecting the data from sensor nodes and transmitting towards the final distribution. At the relay nodes, the data collected by the sensor nodes generally come in batches and the relay nodes transmit those data in batches also rather than individually in order to minimize transmission times. We can model this by bulk arrivals and bulk service queueing models. A major challenge in WSNs is energy consumption of relay as well as sensor nodes, as they are typically powered by batteries with limited lifespans. To reduce energy consumption with preserving the QoS, and for the improvement of the lifespans of batteries, one of the promising method is the bulk service with batch-size-dependent transmission time and vacation period (which is generally called as sleeping periods). Here we consider only one relay node (which is considered as a single server) and corresponding to one relay node, there is only one waiting line with infinite capacity. The data transmission starts only when a minimum number of data packets, say 'a', are waiting and the maximum 'b' packets can be transmitted at a time. Also the transmission time depends on the size of the transmitted data which reduces the number of transmissions, waiting time, extends the efficiency and enhance the performance of nodes. After completing the transmission of a batch of data, if the number of waiting data is less than 'a', then the relay node becomes inactive (i.e., goes for vacation/sleep).

The vacation time (single or multiple) can exactly be used as sleeping modes (single or multiple) by which the energy consumption in WSNs can be reduced and hence increases the lifetime of batteries. By incorporating the queue-length-dependent vacation (i.e., the vacation period depends on the number of waiting data), the vacation duration can be reduced. After returning from vacation, if the number of accumulated data is at least 'N ($\geq b$)', then the relay node starts some preparatory works (this duration is called as set-up time) before starting the transmission process. Otherwise, again it goes for vacation which is considered as multiple vacations. Generally, the transmission times do not follow any particular probability distribution for all permissible batch sizes. To model such scenarios, the consideration of mixed type of service time distributions (i.e., dynamical switchover of service time distributions with the variant batch size) is more relevant. Here we consider that the minimum service threshold 'a'=7, maximum capacity 'b' =16, N=20 and λ=2 and the others parameters are $g_1 = 0.25$, $g_2 = 0.25$, $g_3 = 0.25$, $g_4 = 0.25$ with mean $\bar{g} = 2.50$, so that $G(z) = 0.25(z + z^2 + z^3 + z^4)$. The mean service rates are $\mu_r = \dfrac{8.0}{r+1}$, $7 \leq r \leq 16$. When the server provides

the service to a group of customers with sizes 7, 8, 9 or 10, the service times follow a two-phase hyper-exponential distribution (HE_2) with mean service rate in individual phase are $\mu_{1,r} = \frac{3}{4}\mu_r$, $\mu_{2,r} = \frac{3}{2}\mu_r$, $r = 7, 8, 9, 10$ and the phase selection probabilities are $\sigma_1 = 0.5$, $\sigma_2 = 0.5$, so that $K^{(r)}(z) = \sigma_1 \frac{\mu_{1,r}}{\mu_{1,r} + \lambda - \lambda G(z)} + \sigma_2 \frac{\mu_{2,r}}{\mu_{2,r} + \lambda - \lambda G(z)}$. For the batch sizes 11 to 13, the service times follow Erlang distribution with phase 2 (E_2) and the mean service time of each phase is $\frac{1}{2\mu_r}$, $r = 11, 12, 13$ such that overall mean service time is $\frac{1}{\mu_r}$. Finally, for serving the batch of sizes 14 to 16, the service times follow deterministic distribution (D) and its L.S.T is $e^{-\frac{1}{\mu_r}}$, $14 \le r \le 16$. It is a transcendental function and in order to convert it into a rational function in the form $\frac{P(\theta)}{Q(\theta)}$, we use the Pade [45, 50] approximation. The vacation times follow hypo-exponential distribution (HYE_2) with the rate of k-type vacation is $\eta_k = (k+1) \cdot 0.85$, $i = 0, \ldots, N-1$ such at $M^{[k]}(z) = \left(\frac{\frac{5}{4}\eta_k}{\frac{5}{4}\eta_k + \lambda - \lambda G(z)} \right) \cdot \left(\frac{5\eta_k}{5\eta_k + \lambda - \lambda G(z)} \right)$. Also, the set-up time follows exponential distribution with parameter $\kappa = 2.0$. Due to the lack of space, all the tables containing numerical values of joint probabilities are not displayed in this article, but, can be retrieved from the link mentioned in Section 2.

8.1. Sensitivity analysis of the performance measures

After verifying the obtained theoretical results through the numerical example, in this subsection, we explore the effectiveness of different values of the parameters on the system performance measures subject to the stability condition of the system is preserved. Tables 1–2 present the effect of arrival rate on the performance measures such as average queue length (L_q), average system size (L), average waiting time in queue (W_q), average waiting time in system (W), the probability that the server is on vacation (Q_{vac}), set-up (P_{set}), busy state (P_{busy}) for deterministic (D), exponential (M), Erlang with phase-3 (E_3) distributions as service times in Table 1 and vacation times in Table 2. In Table 1, the vacation and set-up times follow HE_2 and M distribution, respectively, whereas in Table 2, the service and set-up times follow HE_2 and M distribution, respectively. For Tables 1–2, the parameters are chosen as $a = 3$, $b = 7$, $N = 10$, $g_1 = 0.06$, $g_2 = 0.09$, $g_3 = 0.30$, $g_4 = 0.30$, $g_5 = 0.20$, $g_6 = 0.05$, $g_n = 0$, $n \ge 7$, $\mu_r = \frac{12.0}{r}$, $r = a, \ldots, b$, $\eta_k = \frac{(i+1)}{2} \times 0.85$, $\nu = 1.2$, so that $\tilde{S}_r(\theta) = e^{-\frac{\theta}{\mu_r}}$ for D, which is a transcendental function, during computational it has been approximated to a rational function using Pade'(30, 35) approximation. In Tables [1 - 2], the busy probability are same for all distributions but the other system's characteristics crucially varies for different service and vacation times distributions.

From Table 1, we observe that increasing of arrival rate enhances the queue length as well as the system size for both single as well as multiple vacation cases, irrespective of any service times distribution, which is significant and agrees with real life scenarios. Whereas the average waiting times in queue and system are initially decreasing but for $\lambda > 1.4$, the waiting times are gradually increasing for single vacation case and for $\lambda \ge 1.6$, the same scenario occurs for multiple vacations cases. The reason is that initially the server gets an adequate number of customers always which reduces the customer's waiting time but for the higher values of λ, more customers come in a short interval of time which generally leads the system over crowded and eventually increases the waiting time.

The probability that the server is on vacation and set-up state gradually decreases with the increases of arrival rate. The reason behind this is that most of the time, the required number of customers are available to start the service which reduces the vacation rate as well set-up rate. Meanwhile, for a lower value of λ, the difference between different performance measures for D, M and E_3 are not very significant. In fact, its effect can be noticed when $\lambda \ge 1.8$, as the difference in the values of performance measures becomes more significant for the different service time distributions. This means that for the lower arrival rate, the different service time distributions has less impact on the system's performance. Hence it points out that the various service time distributions play a crucial role in determining the performance measures of the system for a moderate to higher value of arrival rate and should be taken into consideration during the modeling of practical circumstances. One noticeable point is that for D service time distribution, L_q, L, W_q and W are less as compared with other distributions. From this, it can be concluded that various service time distributions have significant effects on the system's performance and by proper choice, we can reduce the various system performance measures which eventually improves the QoS and leads to less operating cost. A similar type of observations are shown in Table 2.

For Figs. 5–12, we consider the service, vacation and set-up times as D, HYE_2 and M distribution, respectively, with $a = 3$, $g_1 = 0.06$, $g_2 = 0.09$, $g_3 = 0.30$, $g_4 = 0.30$, $g_5 = 0.20$, $g_6 = 0.05$, $g_n = 0$, $n \ge 7$ and set-up rate $(\nu) = 1.2$. In the Figs. 5–6, the other input parameters are taken as $\lambda = 1.1$, $b = 7$, $\eta_k = \frac{k+1}{2} \times 0.85$ and the service rate (μ_r) are chosen in such a way that the stability condition is preserved, whereas for Figs. 7–8, $b = 17$, $N = 10$ and $\mu_r = \frac{4.24}{r}$ and in Figs. 9–12, $N = 15$, $\mu_r = \frac{9}{r}$ and vacation rate is the same as Fig. 5. The Fig. 5 depicts the impact of ρ and N on Lq for both single and multiple vacations, where N varies from 10 to 16 and ρ varies from 0.2 to 0.8. It is observed from Fig. 5 that L_q increases with the increase of ρ when N is fixed. The reason is that for increasing the value of ρ, the service rate is decreasing for which the server is busy with a batch more times which naturally increased L_q. One significant observation is that for a lower value of ρ, L_q rises gradually. But for $\rho \ge 0.65$, it increases faster. Also, in the case of multiple vacations, L_q is always greater than compared to single vacation due to the frequent vacations. Further, for fixed ρ, L_q increases when N increases. It is obvious that for the higher value of N, once the server goes on vacation then it has to wait more times to resume service, and hence the system becomes overcrowded. Where Fig. 6 illustrates the impact of ρ and N on

Table 1
Effect of arrival rate (λ) on the system performance measures for different service time distributions.

	λ	Single vacation							Multiple vacation						
		L_q	L	W_q	W	Q_{vac}	P_{set}	P_{busy}	L_q	L	W_q	W	Q_{vac}	P_{set}	P_{busy}
Deterministic distribution	0.8	6.576593	8.160452	8.220741	10.200565	0.306088	0.124507	0.242667	7.137567	8.730735	8.921959	10.913419	0.639840	0.117494	0.242667
	1.0	7.712832	9.709088	7.712832	9.709088	0.320241	0.131888	0.303333	8.328683	10.335235	8.328683	10.335235	0.572853	0.123814	0.303333
	1.2	8.994205	11.407196	7.495171	9.505997	0.319899	0.133022	0.364000	9.632035	12.055659	8.026696	10.046382	0.511400	0.124599	0.364000
	1.4	10.405089	13.238712	7.432206	9.456223	0.308827	0.129440	0.424667	11.041992	13.886106	7.887137	9.918647	0.459119	0.121214	0.424667
	1.6	11.943205	15.201082	7.464503	9.500676	0.289872	0.122339	0.485333	12.564284	15.832167	7.852677	9.895105	0.399987	0.114679	0.485333
	1.8	13.623243	17.308836	7.568468	9.616020	0.265137	0.112613	0.546000	14.219423	17.914291	7.899680	9.952384	0.348253	0.105747	0.546000
	2.0	15.484067	19.600772	7.742034	9.800386	0.236161	0.100916	0.606667	16.050289	20.175355	8.025144	10.087677	0.298365	0.094968	0.606667
	2.2	17.605329	22.156547	8.002422	10.071158	0.204065	0.087719	0.667333	18.139205	22.697734	8.245093	10.317152	0.249921	0.082745	0.667333
	2.4	20.148180	25.137382	8.395075	10.473909	0.169668	0.073368	0.728000	20.649053	25.644410	8.603772	10.685171	0.202624	0.069376	0.728000
	2.6	23.468018	28.898798	9.026161	11.114922	0.133572	0.058107	0.788667	23.936314	29.372006	9.206275	11.296925	0.156255	0.055078	0.788667
Exponential distribution	0.8	6.828768	8.417342	8.535960	10.521677	0.306499	0.124437	0.242667	7.390818	8.988807	9.238522	11.236009	0.639859	0.117474	0.242667
	1.0	8.145638	10.150043	8.145638	10.150043	0.320533	0.131824	0.303333	8.761894	10.776564	8.761894	10.776564	0.572847	0.123819	0.303333
	1.2	9.682356	12.107409	8.068630	10.089508	0.320082	0.132971	0.364000	10.319822	12.755258	8.599851	10.629381	0.511369	0.124630	0.364000
	1.4	11.448414	14.298124	8.177440	10.212946	0.308953	0.129397	0.424667	12.084268	14.943964	8.631620	10.674260	0.454074	0.121259	0.424667
	1.6	13.478451	16.756199	8.424032	10.472625	0.290003	0.122291	0.485333	14.097924	17.384927	8.811203	10.865579	0.399947	0.114719	0.485333
	1.8	15.843256	19.551957	8.801809	10.862199	0.265321	0.112542	0.546000	16.437389	20.154396	9.131883	11.196887	0.348238	0.105762	0.546000
	2.0	18.671594	22.813794	9.335797	11.406897	0.236425	0.100807	0.606667	19.235396	23.384832	9.167698	11.692416	0.298391	0.094943	0.606667
	2.2	22.197599	26.775562	10.089817	12.170710	0.204413	0.087566	0.667333	22.728716	27.312782	10.331234	12.414901	0.249996	0.082671	0.667333
	2.4	26.873620	31.889394	11.197342	13.287248	0.170080	0.073173	0.728000	27.371399	32.392119	11.404749	13.496716	0.202747	0.069253	0.728000
	2.6	33.679362	39.134820	12.953601	15.051854	0.134010	0.057887	0.788667	34.144217	39.603470	13.132391	15.232104	0.156413	0.054920	0.788667
Erlang-3 distribution	0.8	6.660294	8.246022	8.325367	10.307528	0.306207	0.124486	0.242667	7.221575	8.816655	9.026969	11.020819	0.639842	0.117491	0.242667
	1.0	7.856153	9.855747	7.856153	9.855747	0.320294	0.131875	0.303333	8.472023	10.481898	8.472023	10.481898	0.572841	0.123826	0.303333
	1.2	9.221640	11.639716	7.684699	9.699763	0.319893	0.133019	0.364000	9.859184	12.287782	8.215987	10.239818	0.511373	0.124627	0.364000
	1.4	10.749546	13.590131	7.678247	9.707236	0.308790	0.129446	0.424667	11.385923	14.236772	8.132802	10.169123	0.454082	0.121252	0.424667
	1.6	12.450033	15.716723	7.781270	9.822952	0.289836	0.122347	0.485333	13.070421	16.346772	8.169013	10.216733	0.399949	0.114718	0.485333
	1.8	14.356679	18.052761	7.975933	10.029312	0.265130	0.112613	0.546000	14.952064	18.656967	8.306702	10.364981	0.348224	0.105776	0.546000
	2.0	16.538528	20.667067	8.269264	10.333533	0.236200	0.100898	0.606667	17.103883	21.240244	8.551942	10.620122	0.298354	0.094979	0.606667
	2.2	19.126958	23.690861	8.694072	10.768573	0.204156	0.087679	0.667333	19.659903	24.230529	8.936320	11.013877	0.249934	0.082733	0.667333
	2.4	22.381614	27.382686	9.325672	11.409452	0.169803	0.073303	0.728000	22.880292	27.887918	9.533455	11.619997	0.202662	0.069338	0.728000
	2.6	26.862656	32.305638	10.331791	12.425246	0.133736	0.058024	0.788667	27.329859	32.777186	10.511484	12.606610	0.156314	0.055019	0.788667

the probability that server is on vacation. For the fixed value of N, the vacation probability decreases with the increase of ρ. The behind cause is that the service rate is decreasing for which the server is getting an adequate number of customers to render the service for which the vacation rate reduces. On the other hand, for a fixed value of ρ, the vacation probability decreases quietly for a larger value of N in a single vacation case. But the different trend is observed for multiple vacations. The reason is that in multiple vacations cases, the server goes for vacation frequently which increases the vacation probability. In a single vacation case, after returning from first vacation period, the server waits a long time in the dormant state and then starts the service which intuitively reduces the vacation rate. For the higher value of ρ, the probability that the server is on vacation almost same and goes to zero for both cases because the server is busy with some customers. Fig. 7 represents the effect of arrival and vacation rate on the average system length, where vacation rate is $\eta_k = \frac{(k+1)}{2} \times \kappa, \kappa = 0.75(0.10)1.25$. It is seen that for a fixed vacation rate, increasing of arrival rate enhances L which is natural, this is due to the more arrival of customers in a short span of time creating over congestion. On the other hand, for a fixed value of λ, when the vacation rate is increasing, L decreases. Due to the higher value of the vacation rate, the server returns from vacation in a short time, for which the server starts service quickly and hence reduces L. It indicates that for a higher arrival rate, we can reduce the overcrowded by proper monitoring of vacation duration. Fig. 8 shows that when λ is fixed and vacation rate is increasing, the probability that the server is in set-up mode is also increasing steadily. This is because with the higher vacation rate, the server comes back more frequently from the vacation state to the set-up state and eventually increases the probability. Also, for a fixed vacation rate, the probability that the server is in set-up state is gradually increasing with the increase of arrival rate because the server commences service with maximum number of customers always for which it goes for vacation and returns on set-up state frequently. In multiple vacations cases, the set-up probability is always less as compared to single vacation as the server is on vacation period more times.

Figs. 9–10 show the impact of λ and b on L_q and L for single and multiple vacations, respectively. When λ increases, L_q and L increase in both the cases as expected. However, for fixed λ, it is showing different tendencies between L_q and L with the increase of 'b'. Since the server initiates service with more customers which reduces the queue length, but more customers with the server which produces a higher value of L. The similar arguments hold for the behavior of W_q and W in Figs. 11–12. It indicates that when the arrival rate is high, the system manager can maintain the traffic by choosing the service capacity to a moderate value so that the average queue/system length and waiting time in the queue/system can be maintained to satisfactory level of customers.

In Figs. 5–12, we have considered only the queue-length-dependent vacation policy (QLDV). Whereas, in Figs. 13–14, a comparison has been drawn between QLDV and queue-length-independent vacation (QLIV) (i.e., the vacation time does not depend on the queue size at vacation initiation epoch) on L_q with ρ for D, M and E_3 service times distributions. For Figs. 13–14, we consider $\lambda = 1.5, \mu_r = \frac{21.84}{r}$ and all other parameters are the same as in Table 1 for the QLDV case. For the QLIV case, all the parameters are the same as previous except η_k which is taken as $0.425,\ 0 \le k \le N-1$. From Figs. 13–14, it is observed that L_q is always less for QLDV as compared to QLIV irrespective of service time distributions. This indicates that QLDV makes the system more efficient and improves QoS than QLIV.

8.2. System cost optimization

From the application point of view of the concerned model, the cost function plays an important role to determine the optimal values of system parameters such that the operating cost of the system is minimal. We employ the same approach/concept given

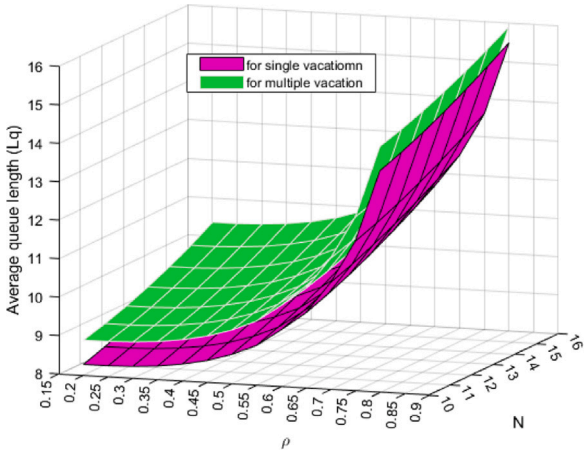

Fig. 5. Average queue length versus ρ versus N. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

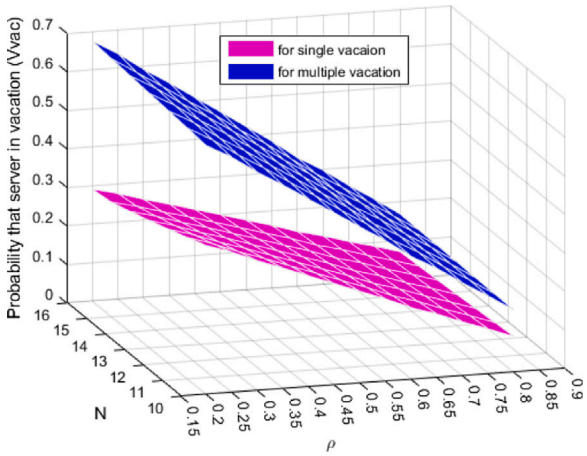

Fig. 6. The probability that server is in vacation versus N versus ρ. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

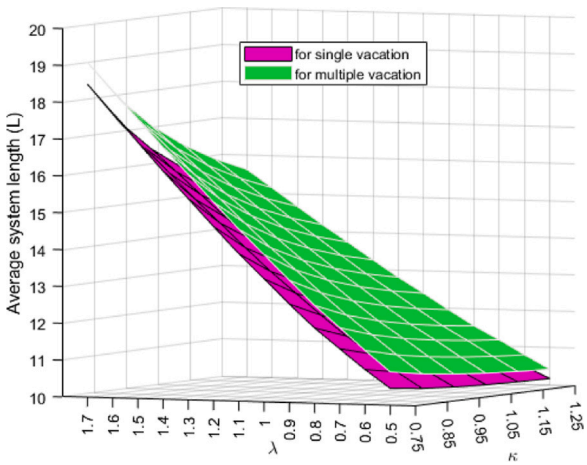

Fig. 7. Average system length versus arrival rate versus vacation rate. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 2
Effect of arrival rate (λ) on the system performance measures for different vacation time distribution.

	λ	Single vacation							Multiple vacation						
		L_q	L	W_q	W	Q_{vac}	P_{set}	P_{busy}	L_q	L	W_q	W	Q_{vac}	P_{set}	P_{busy}
Deterministic distribution	0.8	6.100901	7.682367	7.626127	9.602959	0.319788	0.129718	0.242667	6.339744	7.926506	7.924681	9.908133	0.631136	0.126197	0.242667
	1.0	7.017980	9.012281	7.017980	9.012281	0.338492	0.139195	0.303333	7.262538	9.262412	7.262538	9.262412	0.561393	0.135273	0.303333
	1.2	8.110271	10.522759	6.758559	8.768965	0.340714	0.141640	0.364000	8.341959	10.759777	6.951633	8.966481	0.498227	0.137772	0.364000
	1.4	9.388777	12.224169	6.706289	8.731549	0.330203	0.138493	0.424667	9.596752	12.436915	6.854823	8.883510	0.440348	0.134985	0.424667
	1.6	10.876414	14.138831	6.797759	8.836770	0.310231	0.131084	0.485333	11.056321	14.322808	6.910200	8.951755	0.386577	0.128089	0.485333
	1.8	12.619677	16.312718	7.010932	9.062621	0.283442	0.120529	0.546000	12.771524	16.467911	7.095291	9.148839	0.335914	0.118086	0.546000
	2.0	14.706483	18.833302	7.353241	9.416659	0.251852	0.107700	0.606667	14.832675	18.962176	7.416337	9.481088	0.287555	0.105778	0.606667
	2.2	17.301166	21.864631	7.864167	9.938469	0.216940	0.093245	0.667333	17.405099	21.970627	7.911409	9.986648	0.240884	0.091783	0.667333
	2.4	20.725777	25.728458	8.635740	10.720191	0.179754	0.077629	0.728000	20.810983	25.815205	8.671243	10.756335	0.195442	0.076558	0.728000
	2.6	25.682045	31.126347	9.877709	11.971672	0.141027	0.061178	0.788667	25.751756	31.197153	9.904521	11.998905	0.150901	0.060432	0.788667
Exponential distribution	0.8	7.207763	8.797416	9.009704	10.996770	0.298615	0.121408	0.242667	8.129031	9.731385	10.161289	12.164231	0.645976	0.111357	0.242667
	1.0	8.666077	10.671063	8.666077	10.671063	0.310843	0.127954	0.303333	9.708507	11.727619	9.708507	11.727619	0.580365	0.116302	0.303333
	1.2	10.317637	12.742378	8.598031	10.618648	0.309537	0.128652	0.364000	11.455987	13.875423	9.529989	11.562853	0.519672	0.116327	0.364000
	1.4	12.156379	15.004645	8.683128	10.717604	0.298346	0.124985	0.424667	13.317730	16.180612	9.512664	11.557580	0.462616	0.112717	0.424667
	1.6	14.198873	17.473977	8.874296	10.921235	0.279908	0.118056	0.485333	15.380025	18.669174	9.612515	11.668234	0.408288	0.106378	0.485333
	1.8	16.490481	20.195406	9.161378	11.219670	0.256121	0.108672	0.546000	17.675389	21.393416	9.819661	11.785231	0.356046	0.097954	0.546000
	2.0	19.119956	23.257445	9.559978	11.628722	0.228349	0.097419	0.606667	20.297686	24.447046	10.148843	12.223523	0.305427	0.087906	0.606667
	2.2	22.252861	26.825480	10.114937	12.193400	0.197583	0.084723	0.667333	23.416078	27.999114	10.643672	12.726870	0.256095	0.076572	0.667333
	2.4	26.212241	31.222429	10.921767	13.009345	0.164546	0.070879	0.728000	27.356135	32.375108	11.398389	13.489628	0.207802	0.064198	0.728000
	2.6	31.700441	37.150541	12.192477	12.288669	0.129775	0.056174	0.788667	32.821968	38.279076	12.623834	14.722721	0.160362	0.050972	0.788667
Erlang-3 distribution	0.8	6.465171	8.050167	8.081463	10.062709	0.310929	0.126301	0.242667	6.879646	8.472389	8.599557	10.590486	0.636577	0.120756	0.242667
	1.0	7.581162	9.580278	7.581162	9.580278	0.326248	0.134267	0.303333	8.027501	10.035025	8.027501	10.035025	0.568745	0.127922	0.303333
	1.2	8.880000	11.298142	7.400000	9.415118	0.326442	0.135724	0.364000	9.331586	11.758168	7.776322	9.798473	0.506818	0.129182	0.364000
	1.4	10.363126	13.204513	7.402233	9.431795	0.315379	0.132215	0.424667	10.802175	13.651605	7.715839	9.751146	0.449409	0.125923	0.424667
	1.6	12.049958	15.318277	7.531224	9.573923	0.296066	0.124990	0.485333	12.465784	15.741475	7.791115	9.838422	0.395431	0.119235	0.485333
	1.8	13.986755	17.685274	7.770419	8.825152	0.270747	0.115002	0.546000	14.373714	18.078779	7.985397	10.043766	0.344056	0.109944	0.546000
	2.0	16.262455	20.394129	8.131227	10.197065	0.241064	0.102959	0.606667	16.618242	20.755556	8.309121	10.377778	0.294666	0.098667	0.606667
	2.2	19.042628	23.610172	8.655740	10.731896	0.208208	0.089379	0.667333	19.367015	23.939268	8.803189	10.881485	0.246802	0.0858365	0.667333
	2.4	22.650317	27.656271	9.437632	11.523446	0.173037	0.074637	0.728000	22.944311	27.954043	9.560129	11.647518	0.200124	0.071876	0.728000
	2.6	27.787914	33.234683	10.687659	12.782570	0.136173	0.059005	0.788667	28.053189	33.502831	10.789688	12.885704	0.154379	0.056954	0.788667

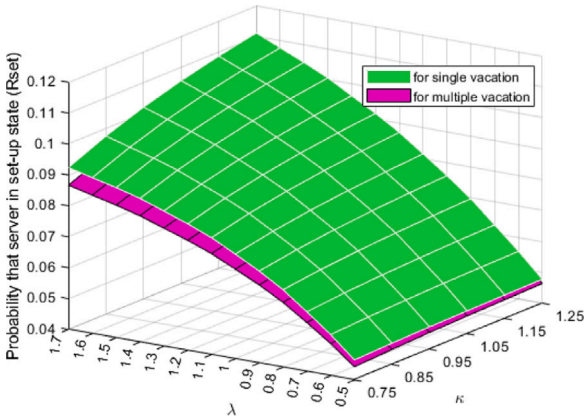

Fig. 8. The probability that server is in set-up state versus arrival rate versus vacation rate. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

in Wang et al. [34,35], Yang and Ke [36], for the construction of the cost function. Additionally, we also incorporate start-up cost and shutdown cost (only once during a complete cycle length) using the concept of Tadj and Choudhury [37] (page 376). Moreover the cost due to idle period comprises of cost due to dormant period, vacation period and set-up time (in case of single vacation) and in case of multiple vacations, it comprises the vacation period and set-up time. First, we consider the cost parameters as: C_{sc}: server start-up cost per cycle, C_{dc}: server shutdown cost, C_{ic}: cost due to idle period per unit time, C_{rc}: reward due to vacation per unit time, C_{hc}: holding cost per customer, C_{oc}: operating cost per unit time. Now, applying the renewal reward theorem, we get the total cost (TC^*) per unit which is given by:

$$\begin{aligned} TC^* = & \frac{C_{sc}}{E[L_c]} + \frac{C_{dc}}{E[L_c]} + C_{ic}\left[(1-\delta_p)\frac{E[D]}{E[L_c]} + (1-\delta_p)\frac{E[I_1]}{E[L_c]} + \delta_p\frac{E[I_2]}{E[L_c]} + \frac{E[R]}{E[L_c]}\right] + C_{hc}L_q \\ & - C_{rc}\left[(1-\delta_p)\frac{E[I_1]}{E[L_c]} + \delta_p\frac{E[I_2]}{E[L_c]}\right] + \frac{C_{oc}E[B]}{E[L_c]} \end{aligned} \tag{85}$$

where $E[L_c]$ is the expected length of a cycle which is the sum of the expected length of idle and busy periods, i.e., $E[L_c] = E[I]+E[B]$. The derivation of these quantities is elaborately discussed in Section 7.1 and each cycle length along with renewal points are shown in Figs. 3–4. It is noteworthy to mention that the cost function exhibits a high degree of complexity and nonlinearity, likely due to the presence of numerous parameters. So it is difficult to find the optimal value of parameters analytically. Here we use the direct search method to find out the value of minimum batch size (a) and service rate (μ_r) such that the cost will be minimum. The resulted numerical data can be accessed through the link provided inSection 2. https://github.com/KaranPrasenjit/Supplement. In the supplementary material, Table 5 displays the effect of different values of minimum batch size (a) on TC^* for the fixed value

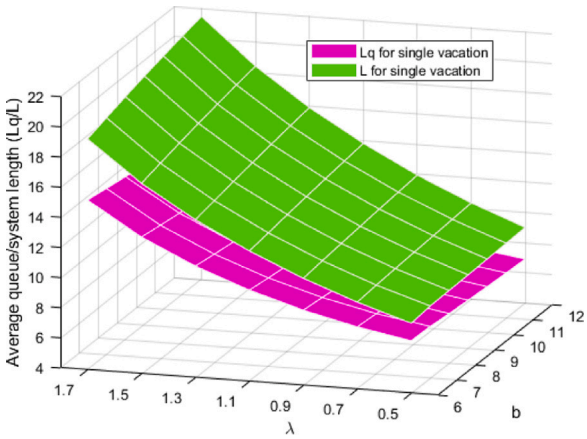

Fig. 9. Average queue/system length versus arrival rate versus b for single vacation. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

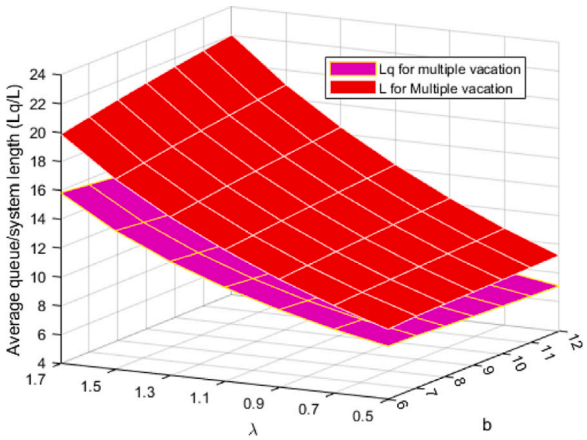

Fig. 10. Average queue/system length versus arrival rate versus b for multiple vacation. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

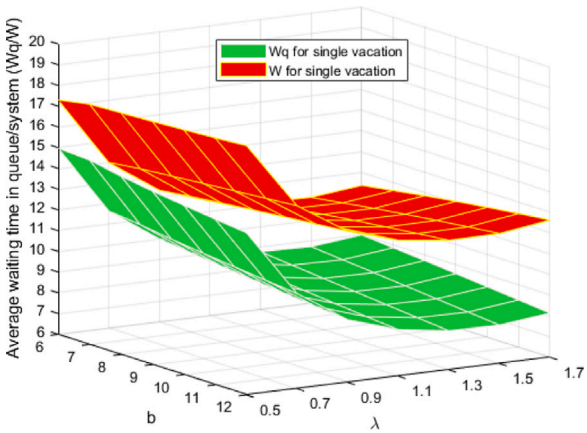

Fig. 11. Average waiting time in queue/system versus b versus arrival rate for single vacation. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

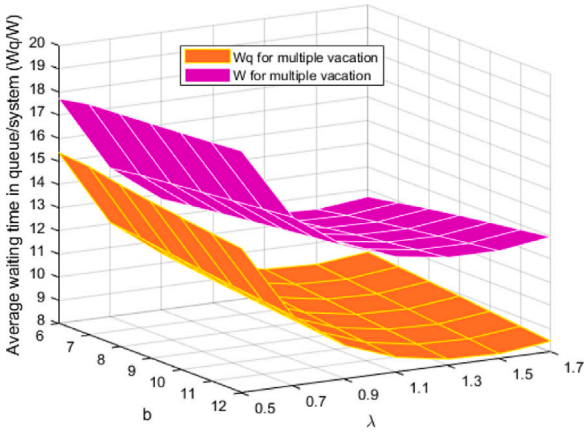

Fig. 12. Average waiting in queue/system versus b versus arrival rate for multiple vacation. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

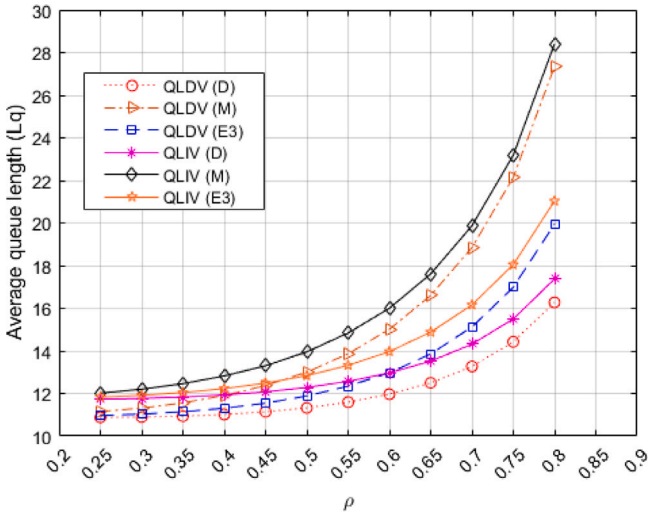

Fig. 13. Average queue length versus traffic intensity for single vacation.

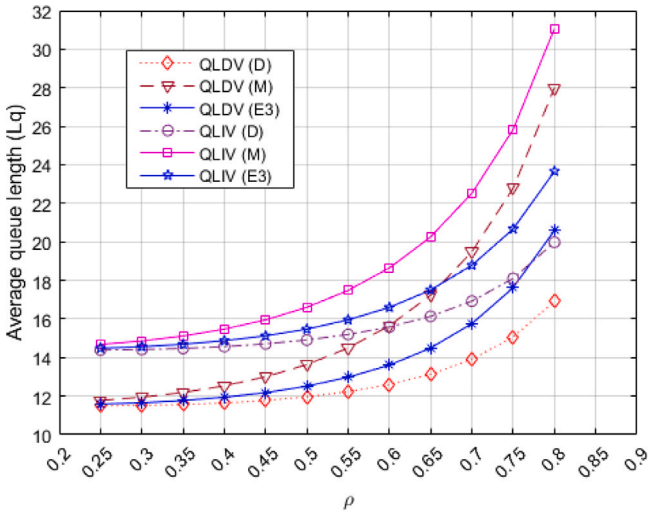

Fig. 14. Average queue length versus traffic intensity for multiple vacations.

of maximum service capacity (b). For this, the value of cost parameters is taken as $C_{sc} = \$5.0$, $C_{dc} = \$\ 3.0$, $C_{ic} = \$1.50$, $C_{rc} = \$\ 1.0$, $C_{hc} = \$0.50$ and $C_{oc} = \$5.0$. The service, vacation and set-up time follow E_2, HYE_2 and M with parameters $\mu_r = \frac{27.5}{r+1}$, $a \le r \le b$, $\eta_k = (k+1) \times 0.85$, $0 \le k \le N-1$ and $\nu = 3.5$, respectively with $b = 12$, $N = 15$, $g_1 = 0.06$, $g_2 = 0.09$, $g_3 = 0.30$, $g_4 = 0.30$, $g_5 = 0.05$, $g_6 = 0.20$ and $g_i = 0, \quad i \ge 7$. From Table 5, it is clear that if the management decides the minimum threshold value as 4 and 2 for single and multiple vacations, respectively, then the operating cost is minimum. For Table 6 in supplementary material, we consider the value of 'a' as 4 and 2 for single and multiple vacations, respectively and all other parameters are the same except the service rate (μ_r) and $\mu_r = \frac{\alpha}{r+1}$, $\alpha = 23.5$ (2) 51.5, $a \le r \le b$. From Tables [5 - 6] in supplementary material, it is clear that if we take $a = 4$ and $\alpha = 45.5$, the operating cost is minimum for a single vacation case. Whereas, the cost is minimal in the case of multiple vacations for $a = 2$ and $\alpha = 43.5$. But if we change the value of cost and other system parameters, then the value of minimum batch size (a) and service rate (μ_r) also change. Furthermore, using the same cost function, we can find out the optimal value of other parameters for minimizing the operating cost.

9. Conclusions and future scopes

In this paper, we have investigated the steady-state behavior of a continuous-time infinite-buffer bulk arrival and batch-size-dependent service queue with queue-length-dependent single as well as multiple vacation (s), N-threshold policy and set-up time. We have provided a simple mechanism to find out the stationary joint distributions at departure and vacation termination epochs using the roots of corresponding characteristic equation of the bivariate probability generating function of queue content and size of departing batch. After deriving the joint distribution, we have obtained a simple relation between departure and arbitrary epochs probabilities. We have calculated the marginal distributions and some significant performance measures. The obtained analytical results have been verified through numerical example. Further, we have presented the impact of the system parameters on the performance measures through tables and graphical presentations which may be helpful to system managers in making some decisions on management as well as economic purposes. Finally, we have constructed a cost function and have determined the optimal value of system parameters to minimize the operating cost of the system per unit of time.

In the future, using the methodologies presented in this current work, anyone can extend this work for more general arrival process likes renewal/non-renewal instead of compound Poisson process and also it can be extended with optional service, single/multiple working vacations, and by allowing the repeated customers etc. Using the cost function addressed in this work, we can find out the optimal value of other system parameters for which the system cost will be the minimum. Some discrete-time models can also be studied using the described methodologies.

CRediT authorship contribution statement

P. Karan: Writing – review & editing, Writing – original draft, Formal analysis, Conceptualization. **S. Pradhan:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Sourav Pradhan reports was provided by India Ministry of Science & Technology Department of Science and Technology. NA If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.peva.2024.102459.

Data availability

No data was used for the research described in the article.

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